



Reference Protocol Guide

This manual supports the following products:

- Revolution™ Maxima

5987224-1EN

Revision: 1

Reference Protocol Guide, English
General User Documentation

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Revision History

Revision History

REV	DATE	REASON FOR CHANGE
1	April 2025	Initial Release

Purpose

Purpose

This document provides a listing of the parameters provided in the reference protocols contained under the GE tab selector. The protocols in Adult are based on parameters needed to scan an average sized adult and take the clinical aim of the protocol into consideration. The protocols in Pediatric are based on age in the Head and Orbit area, and in the rest of the body, use Color Coding for Kids or the Feather Light patient sizing scheme. These protocols are designed as a starting point. They should be reviewed with your Radiologist, Physicist, and Radiation Safety Officer and revised as needed to meet the clinical needs of your department. The document is divided into protocol categories available on the system. Each table lists the Protocol Number, Protocol Name, Post Processing software associated with, Scan Type, SFOV, Pitch/Table Speed/Row, Gantry Speed, Slice Thickness, kV, mA/AVG mA/Min-Max mA/Noise Index, CTDIvol, Recon Algorithm Type used and Phantom type used. Manual mA mode allows you to scan without enabling Auto mA mode. When building protocols, make sure the mA value field contains a reasonable mA value in the event that AutomA is turned off. The Average (Avg) mA has been provided in the tables where Auto mA or Smart mA is used in the protocol so that the dose can be calculated while building protocols. The Avg mA value provided in the table is an estimate based on an average patient size (32- 36 cm). The actual average mA for each image is calculated by using all of the mA values for each scan rotation, as determined by Auto mA for the patient. The Auto mA Theory section in the Learning and Reference Guide, Building Protocols Chapter and in the Technical Reference Manual, General Information Chapter details how Auto mA does this calculation.

Reference Adult Protocols 64 Slice System, 72kW with 0.35sec/ASiR-V

Head

Table 2-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	100	50	STND	AR50	35.81	214.86	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	50	50	STND	AR50	17.90	143.24	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	200	50	STND	AR50	35.81	501.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	STND PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=200	50	STND PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.6	Hi Res Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	DETL PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.7	Hi Res CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=200	50	DETL PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.9	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.10	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.11	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical-S	Head	0.984 39.37 18 pass	0.4	5	40	120	150	50	STND PLUS	AR50	231.41	2892.60	11.00	Head 16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.14	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.15	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 2-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	75	50	STND PLUS IQE+	AR50	25.28	231.40	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	90	50	STND PLUS IQE+	AR40	18.20	143.92	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	85	50	STND PLUS IQE+	AR40	40.49	226.41	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 2-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	150	50	STND	AR70	17.41	243.78	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	120	50	STND	AR70	9.78	78.24	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-500mA NI=12.6 Avg mA=155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-500mA NI=12.6 Avg mA=155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=205	50	STND PLUS IQE+	AR40	9.52	195.59	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=205	50	STND PLUS IQE+	AR40	9.52	195.59	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Hi Res Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep	Prep	Helical	SmallBody	0.516 20.62	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	DETL PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.12	Hi Res Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep	Prep	Helical	SmallBody	0.984 39.37	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=205	50	DETL PLUS IQE+	AR40	9.52	195.59	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.14	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.15	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.16	Hi Res Carotid/ CoW 0.625mm SmartmA/ SmartPrep	Prep	Helical	SmallBody	0.516 20.62	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=125	50	DETL PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.18	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.6	2.5	40	120	SmartmA 80-500mA NI=9.10 Avg mA=105	50	STND PLUS	AR50	4.87	90.42	15.00	Body 32	Neck 2.5mm and SmartmA, Evaluation of soft tissue structures for abnormality
23.19	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=12.60 Avg mA=165	50	STND PLUS IQE+	AR40	7.66	142.09	15.00	Body 32	Neck with DMPR and SmartmA, Evaluation for soft tissue abnormalities

Upper Extremity

Table 2-4
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR and Smart mA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 1.25mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 2-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	98.85	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	55.00 1.375	0.6	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=105	50	STND PLUS	AR40	3.99	98.85	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-500mA NI=12 Avg mA=105	50	CHEST PLUS	AR50	4.27	96.13	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	BONE+	AR50	2.54	60.92	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	118.80	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	5.84	134.60	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	61.25 1.531	0.8	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=125	50	STND PLUS	AR40	5.69	286.81	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	240	50	STND	AR50	12.74	274.59	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=21.45 Avg mA=240	50	STND	AR50	12.74	274.59	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=180	50	STND PLUS	AR40	9.56	320.60	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	STND PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Hi Res Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=180	50	DETL PLUS	AR40	9.56	320.60	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.5:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection.
25.17	Hi Res Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	DETL PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection.
25.19	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	110	50	STND PLUS	AR50	4.18	103.56	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	50	50	STND	AR50	28.01	448.09	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.20	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.22	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375 55.0 11 pass	0.4	5	40	120	SmartmA 50-500mA NI=14 Avg mA=175	50	STND PLUS	AR50	61.64	1941.68	30.00	Body 32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature
25.24	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	32.04	1073.36	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.25	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	215	50	STND SEGM	AR50	4.84	48.45	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.27	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	32.16	916.53	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	39.37 0.984	0.35	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=125	50	STND PLUS	AR40	3.87	207.17	49.94	Body 32	
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.33	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.34	SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	275	50	STND SSSEG	AR40	12.04	168.63	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
25.35	Hi Res SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSSEG	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30- 74 bpm
25.36	Hi Res SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.37	Hi Res SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.38	Hi Res SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	275	50	DETL SSSEG	AR40	12.04	168.63	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
25.40	SnapShot Assist Coronary		Helical	LargeBody	55.0 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries with SnapShot Assist.
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries with SnapShot Assist.
			Cardiac SEG	CardiacLarge	6.4 0.16	0.35	0.625	40	120	ECG 250-450mA Phase 70-80 Avg mA=450	0	STND SSSEG	NONE	66.11	892.54	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries with SnapShot Assist.
25.41	SnapShot Assist Function		Helical	LargeBody	55.0 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of cardiac function with SnapShot Assist.
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of cardiac function with SnapShot Assist.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	6.4 0.16	0.35	0.625	40	120	450	0	STND SSSEG	NONE	85.74	1157.48	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of cardiac function with SnapShot Assist.

Abdomen

Table 2-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-500mA NI=18 Avg mA=195	50	STND PLUS	AR40	13.80	601.03	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.4	Hi Res Abdomen Pelvis +DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-500mA NI=18 Avg mA=195	50	DETL PLUS	AR40	13.80	601.03	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.6	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.7	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.9	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	140	50	STND PLUS	AR40	10.81	470.63	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.10	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=140	50	STND PLUS	AR40	10.81	470.63	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Hi Res AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	140	50	DETL PLUS	AR40	10.81	470.63	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.12	Hi Res AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=140	50	DETL PLUS	AR40	10.81	470.63	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.14	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.15	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.16	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	189.95	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-500mA NI=15 Avg mA=150	50	STND PLUS	AR50	7.96	107.89	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	222.64	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.17	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.18	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.20	VHS Liver Perfusion		Helical	LargeBody	39.37 0.984	0.6	5	40	120	85	50	STND PLUS	AR50	4.51	74.68	13.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		Perfusion	Helical-S/P	LargeBody	55.0 1.375 21 pass	0.4	5	40	120	150	50	STND PLUS	AR50	126.99	1714.31	12.00	Body 32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.21	VHS 4D CTA		Helical	LargeBody	39.37 0.984	0.6	5	40	120	85	50	STND PLUS	AR50	4.51	74.68	13.00	Body 32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	STND	NONE	118.40	59.20	0.00	Body 32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	55.0 1.375 18 pass	0.4	5	40	120	SmartmA 50-500mA NI=13 Avg mA=175	50	STND PLUS	AR50	112.73	2198.32	18.00	Body 32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.23	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	175	50	STND PLUS	AR40	13.51	1398.84	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.25	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=175	50	STND PLUS	AR40	13.51	1398.84	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.26	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.27	Hi Res Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=175	50	DETL PLUS	AR40	13.51	1398.84	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.28	Hi Res Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=125	50	DETL PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.30	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 2-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	90	50	STND	AR60	24.39	48.78	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-440mA NI=14 Avg mA=130	50	BONE PLUS	AR50	25.54	534.75	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-440mA NI=20 Avg mA=180	50	BONE PLUS IQE+	AR50	44.21	925.44	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-440mA NI=18 Avg mA=90	50	BONE PLUS IQE	AR50	22.11	462.69	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-440mA NI=20 Avg mA=180	50	BONE PLUS IQE+	AR50	44.21	925.44	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 2-8
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	BONE PLUS IQE	AR50	14.60	199.44	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	8.85	252.55	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 2-9
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	60	50	STND PLUS	AR50	10.14	95.59	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	80	50	STND PLUS IQE+	AR40	14.03	73.22	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	60	50	STND PLUS IQE+	AR40	10.53	54.92	3.50	Body 32	Helical scan mode 0.625mm , Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 64 Slice System, 72kW with 0.35sec/ ASiR-V

Head

Table 3-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	145	0	STND	AR40	12.98	155.77	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	200	0	STND	AR40	17.90	214.86	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	110	0	STND	AR40	19.70	236.34	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 3-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	AR20	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	90	0	STND PLUS IQE+	AR40	15.17	119.96	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	85	0	STND PLUS IQE+	AR40	14.32	80.17	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 3-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	50	0	STND PLUS	AR40	1.19	11.88	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSSEG	AR30	8.00	88.01	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSB	AR30	8.00	88.01	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 35-85 Avg mA=170	0	STND SSB	AR30	8.57	94.27	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-200mA NI=18 Avg mA=120	40	STND PLUS	AR30	1.24	13.75	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	AR40	1.66	16.64	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSSEG	AR30	8.45	92.91	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSB	AR30	8.45	92.91	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 35-85 Avg mA=180	0	STND SSB	AR30	9.06	99.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-220mA NI=18 Avg mA=130	40	STND PLUS	AR30	1.35	14.90	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBo dy	27.50 1.375	0.6	3.75	20	120	60	0	STND PLUS	AR40	2.13	22.16	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSSEG	AR30	8.98	98.76	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSB	AR30	8.98	98.76	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 35-85 Avg mA=190	0	STND SSB	AR30	9.59	105.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-240mA NI=18 Avg mA=145	40	STND PLUS	AR30	1.50	16.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	55	0	STND PLUS	AR40	1.96	22.52	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSSEG	AR30	9.65	106.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSB	AR30	9.65	106.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 35-85 Avg mA=205	0	STND SSB	AR30	10.34	113.72	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-280mA NI=18 Avg mA=150	40	STND PLUS	AR30	1.55	17.19	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	60	0	STND PLUS	AR40	2.13	24.56	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	10.77	118.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	10.77	118.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	11.58	127.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-300mA NI=18 Avg mA=180	40	STND PLUS	AR30	1.86	20.63	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	AR40	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	AR30	11.23	123.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSB	AR30	11.23	123.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 35-85 Avg mA=240	0	STND SSB	AR30	12.08	132.87	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 100-340mA NI=18 Avg mA=205	40	STND PLUS	AR30	2.12	23.50	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	65	0	STND PLUS	AR40	2.47	36.49	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	AR30	18.64	205.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSB	AR30	18.64	205.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 35-85 Avg mA=210	0	STND SSB	AR30	20.09	220.94	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.4	0.625	40	100	SmartmA 100-360mA NI=18 Avg mA=205	40	STND PLUS	AR30	4.04	44.71	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	70	0	STND PLUS	AR40	2.66	44.62	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	20.50	225.47	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	20.50	225.47	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	22.03	242.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-400mA NI=16 Avg mA=240	40	STND PLUS	AR30	5.91	65.37	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	80	0	STND PLUS	AR40	3.04	60.11	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSSEG	AR30	22.85	251.32	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSB	AR30	22.85	251.32	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 35-85 Avg mA=250	0	STND SSB	AR30	24.20	266.21	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-440mA NI=16 Avg mA=240	40	STND PLUS	AR30	5.91	65.37	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 3-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	60	0	STND PLUS	AR40	1.42	14.79	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	95	0	STND PLUS	AR40	2.25	25.95	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	AR40	2.61	32.99	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	75	0	STND PLUS	AR40	2.22	30.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	80	0	STND PLUS	AR40	2.37	32.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	LargeBody	27.50 1.375	0.5	5	20	120	70	0	STND PLUS	AR40	2.37	36.81	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	90	0	STND PLUS	AR40	2.85	56.38	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	100	0	STND PLUS	AR40	3.17	62.65	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	AR40	3.80	94.18	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 3-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 64 Slice System, 55kW with 0.35sec/ASiR-V

Head

Table 4-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	100	50	STND	AR50	35.81	214.86	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	50	50	STND	AR50	17.90	143.24	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	200	50	STND	AR50	35.81	501.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	STND PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=200	50	STND PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.6	Hi Res Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	DETL PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.7	Hi Res CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=200	50	DETL PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.9	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.10	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.11	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical-S	Head	0.984 39.37 18 pass	0.5	5	40	120	120	50	STND PLUS	AR50	208.65	2608.18	11.00	Head 16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.14	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.15	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 4-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	75	50	STND PLUS IQE+	AR50	25.28	231.40	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	90	50	STND PLUS IQE+	AR40	18.20	143.92	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	85	50	STND PLUS IQE+	AR40	40.49	226.41	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 4-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	150	50	STND	AR70	17.41	243.78	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	120	50	STND	AR70	9.78	78.24	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-400mA NI=12.6 Avg mA=155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-400mA NI=12.6 Avg mA=155	50	BONE PLUS IQE+	AR50	18.32	264.39	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=95	50	STND PLUS IQE+	AR40	11.23	229.41	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA, Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=155	50	STND PLUS IQE+	AR40	9.59	197.09	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=95	50	STND PLUS IQE+	AR40	11.23	229.41	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=155	50	STND PLUS IQE+	AR40	9.59	197.09	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Hi Res Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=95	50	DETL PLUS IQE+	AR40	11.23	229.41	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.12	Hi Res Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=155	50	DETL PLUS IQE+	AR40	9.59	197.09	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.14	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.15	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=125	50	STND PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.16	Hi Res Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=125	50	DETL PLUS IQE+	AR40	11.08	226.43	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.18	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.8	2.5	40	120	SmartmA 80-400mA NI=9.1 Avg mA=80	50	STND PLUS	AR50	4.95	91.82	15.00	Body 32	Neck 2.5mm and SmartmA, Evaluation of soft tissue structures for abnormality
23.19	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=12.6 Avg mA=125	50	STND PLUS IQE+	AR40	7.74	143.47	15.00	Body 32	Neck with DMPR and SmartmA, Evaluation for soft tissue abnormalities

Upper Extremity

Table 4-4
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 2.5mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 2.5mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 2.5mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 4-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	98.85	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	55.00 1.375	0.6	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=105	50	STND PLUS	AR40	3.99	98.85	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-400mA NI=12 Avg mA=105	50	CHEST PLUS	AR50	4.27	96.13	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	BONE+	AR50	2.54	60.92	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	118.80	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	5.84	134.60	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	61.25 1.531	0.8	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=125	50	STND PLUS	AR40	5.69	286.81	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	180	50	STND	AR50	12.74	274.48	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	SmartmA 50-400mA NI=21.45 Avg mA=180	50	STND	AR50	12.74	274.48	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=180	50	STND PLUS	AR40	9.56	320.60	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	STND PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Hi Res Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=180	50	DETL PLUS	AR40	9.56	320.60	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.5:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.17	Hi Res Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	DETL PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.19	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	110	50	STND PLUS	AR50	4.18	103.56	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	50	50	STND	AR50	28.01	448.09	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.20	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.22	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	55.00 1.375 11 pass	0.5	5	40	120	SmartmA 50-400mA NI=14 Avg mA=140	50	STND PLUS	AR50	57.22	1802.55	30.00	Body 32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature
25.24	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	32.04	1073.36	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.25	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	215	0	STND SEGM	AR50	4.84	48.45	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.27	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	32.16	916.53	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	39.37 0.984	0.35	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=125	50	STND PLUS	AR40	3.87	207.17	49.94	Body 32	
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.33	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.34	SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	275	0	STND SSSEG	AR40	12.04	168.63	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
25.35	Hi Res SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSSEG	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.36	Hi Res SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.37	Hi Res SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSB	AR40	33.04	446.04	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.38	Hi Res SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	275	0	DETL SSSEG	AR40	12.04	168.63	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm

Abdomen

Table 4-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/ MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-400mA NI=18 Avg mA=195	50	STND PLUS	AR40	13.80	601.03	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.4	Hi Res Abdomen Pelvis +DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-400mA NI=18 Avg mA=195	50	DETL PLUS	AR40	13.80	601.03	40.00	Body 32	Basic Abdomen/ Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.6	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.7	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.9	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	105	50	STND PLUS	AR40	10.81	470.56	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
26.10	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=105	50	STND PLUS	AR40	10.81	470.56	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Hi Res AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	105	50	DETL PLUS	AR40	10.81	470.56	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.12	Hi Res AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=105	50	DETL PLUS	AR40	10.81	470.56	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.14	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.15	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.16	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	189.95	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-400mA NI=15 Avg mA=150	50	STND PLUS	AR50	7.96	107.89	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	222.64	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.17	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.18	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.20	VHS Liver Perfusion		Helical	LargeBody	39.37 0.984	0.6	5	40	120	85	50	STND PLUS	AR50	4.51	74.68	13.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		Perfusion	Helical-S/P	LargeBody	55.00 1.375 21 pass	0.5	5	40	120	120	50	STND PLUS	AR50	110.15	1487.04	12.00	Body 32	Volume Helical scan mode for evaluation of liver tumor perfusion over time Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.21	VHS 4D CTA		Helical	LargeBody	39.37 0.984	0.6	5	40	120	85	50	STND PLUS	AR50	4.51	74.68	13.00	Body 32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	STND	NONE	118.40	59.20	0	Body 32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	55.00 1.375 18 pass	0.5	5	40	120	SmartmA 50-400mA NI=13 Avg mA=140	50	STND PLUS	AR50	101.07	1970.92	18.00	Body 32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.23	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	170	50	STND PLUS	AR40	13.12	1358.85	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.25	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=170	50	STND PLUS	AR40	13.12	1358.85	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.26	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
26.27	Hi Res Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=170	50	DETL PLUS	AR40	13.12	1358.85	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, feumrs and lower extremities (Runoff)
26.28	Hi Res Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=125	50	DETL PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evvaluation of vasucular structures of abdomen, femurs and lower extremities (Runoff)
26.30	CT Colonograp hy 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 4-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	90	50	STND	AR60	24.39	48.78	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-340mA NI=14 Avg mA=130	50	BONE PLUS	AR50	25.54	534.75	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-340mA NI=20 Avg mA=170	50	BONE PLUS IQE+	AR50	41.75	873.96	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-340mA NI=18 Avg mA=90	50	BONE PLUS IQE	AR50	22.11	462.69	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-340mA NI=20 Avg mA=170	50	BONE PLUS IQE+	AR50	41.75	873.96	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 4-8
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	BONE PLUS IQE	AR50	14.60	199.44	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	8.85	252.55	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 4-9
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	60	50	STND PLUS	AR50	10.14	95.59	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	80	50	STND PLUS IQE+	AR40	14.03	73.22	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	60	50	STND PLUS IQE+	AR40	10.53	54.92	3.50	Body 32	Helical scan mode 0.625mm , Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 64 Slice System, 55kW with 0.35sec/ ASiR-V

Head

Table 5-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	145	0	STND	AR40	12.98	155.77	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	200	0	STND	AR40	17.90	214.86	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	110	0	STND	AR40	19.70	236.34	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 5-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	AR20	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	90	0	STND PLUS IQE+	AR40	15.17	119.96	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	85	0	STND PLUS IQE+	AR40	14.32	80.17	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 5-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	50	0	STND PLUS	AR40	1.19	11.88	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSSEG	AR30	8.00	88.01	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSB	AR30	8.00	88.01	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-170mA Phase 35-85 Avg mA=170	0	STND SSB	AR30	8.57	94.27	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-200mA NI=18 Avg mA=120	40	STND PLUS	AR30	1.24	13.75	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	AR40	1.66	16.64	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSSEG	AR30	8.45	92.91	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSB	AR30	8.45	92.91	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 50-180mA Phase 35-85 Avg mA=180	0	STND SSB	AR30	9.06	99.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-220mA NI=18 Avg mA=130	40	STND PLUS	AR30	1.35	14.90	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	60	0	STND PLUS	AR40	2.13	22.16	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSSEG	AR30	8.98	98.76	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSB	AR30	8.98	98.76	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-190mA Phase 35-85 Avg mA=190	0	STND SSB	AR30	9.59	105.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-240mA NI=18 Avg mA=145	40	STND PLUS	AR30	1.50	16.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	55	0	STND PLUS	AR40	1.96	22.52	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSSEG	AR30	9.65	106.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSB	AR30	9.65	106.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-205mA Phase 35-85 Avg mA=205	0	STND SSB	AR30	10.34	113.72	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-280mA NI=18 Avg mA=150	40	STND PLUS	AR30	1.55	17.19	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	60	0	STND PLUS	AR40	2.13	24.56	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	10.77	118.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	10.77	118.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	11.58	127.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-300mA NI=18 Avg mA=180	40	STND PLUS	AR30	1.86	20.63	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	AR40	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	AR30	11.23	123.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSB	AR30	11.23	123.48	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 60-240mA Phase 35-85 Avg mA=240	0	STND SSB	AR30	12.08	132.87	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 100-340mA NI=18 Avg mA=205	40	STND PLUS	AR30	2.12	23.50	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	65	0	STND PLUS	AR40	2.47	36.49	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	AR30	18.64	205.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSB	AR30	18.64	205.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 50-210mA Phase 35-85 Avg mA=210	0	STND SSB	AR30	20.09	220.94	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.4	0.625	40	100	SmartmA 100-360mA NI=18 Avg mA=205	40	STND PLUS	AR30	4.04	44.71	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	70	0	STND PLUS	AR40	2.66	44.62	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	20.50	225.47	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	20.50	225.47	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	22.03	242.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-400mA NI=16 Avg mA=240	40	STND PLUS	AR30	5.91	65.37	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	80	0	STND PLUS	AR40	3.04	60.11	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSSEG	AR30	22.85	251.32	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75- 104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSB	AR30	22.85	251.32	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 100-250mA Phase 35-85 Avg mA=250	0	STND SSB	AR30	24.20	266.21	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBo dy	39.37 0.984	0.5	0.625	40	100	SmartmA 100-440mA NI=16 Avg mA=240	40	STND PLUS	AR30	5.91	65.37	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 5-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	60	0	STND PLUS	AR40	1.42	14.79	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	95	0	STND PLUS	AR40	2.25	25.95	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5- 11.5kg (20.9- 25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	AR40	2.61	32.99	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5- 14.5kg (25.4- 32.0lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	75	0	STND PLUS	AR40	2.22	30.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5- 18.5kg (32.0- 40.8lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	80	0	STND PLUS	AR40	2.37	32.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5- 22.5kg (40.8- 49.6lbs)		Helical	LargeBo dy	27.50 1.375	0.5	5	20	120	70	0	STND PLUS	AR40	2.37	36.81	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	90	0	STND PLUS	AR40	2.85	56.38	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	100	0	STND PLUS	AR40	3.17	62.65	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	AR40	3.80	94.18	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 5-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 64 Slice System, 72kW with 0.35sec/ASiR

Head

Table 6-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	120	40	STND	SS40:Slice	42.97	257.83	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	60	40	STND	SS40:Slice	21.49	171.88	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	240	40	STND	SS40:Slice	43.24	605.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	120	40	STND PLUS IQE+	SS40:Slice	24.27	235.88	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=240	40	STND PLUS IQE+	SS40:Slice	48.83	915.01	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.7	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.8	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.9	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical-S	Head	0.984 39.37 18 pass	0.4	5	40	120	180	40	STND PLUS	SS40	277.69	3471.13	11.00	Head 16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.11	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.12	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.13	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 6-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

ProtocolNumber	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	90	40	STND PLUS IQE+	SS40:Slice	30.33	277.68	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	22.24	175.90	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	100	40	STND PLUS IQE+	SS40:Slice	47.63	266.36	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 6-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	185	40	STND	SS40:Slice	21.63	302.85	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	145	40	STND	SS40:Slice	11.82	94.54	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-500mA NI=12.6 Avg mA=185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-500mA NI=12.6 Avg mA=185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA.Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=250	40	STND PLUS IQE+	SS40:Slice	11.61	238.52	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=250	40	STND PLUS IQE+	SS40:Slice	11.61	238.52	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.12	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-500mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.14	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.6	2.5	40	120	SmartmA 80-500mA NI=9.1 Avg mA=125	40	STND PLUS	SS40:Slice	5.80	107.64	15.00	Body 32	Neck 2.5mm and SmartmA. Evaluation of soft tissue structures for abnormality
23.15	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	39.37 0.984	0.6	0.625	40	120	SmartmA 80-500mA NI=12.6 Avg mA=200	40	STND PLUS IQE+	SS40:Slice	9.29	172.23	15.00	Body 32	Neck with DMPR and SmartmA. Evaluation for soft tissue abnormalities

Upper Extremity

Table 6-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR and Smart mA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 1.25mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 6-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	117.68	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	55.00 1.375	0.6	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	117.68	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-500mA NI=12 Avg mA=125	40	CHEST PLUS	SS40:Slice	5.08	114.44	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	230	40	BONE+	SS40:Slice	3.10	74.28	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	141.43	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Slice	6.90	159.07	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	61.25 1.531	0.8	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=155	40	STND PLUS	SS40:Slice	7.05	355.64	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	290	40	STND FULL	SS40:Slice	15.40	331.80	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=21.45 Avg mA=290	40	STND FULL	SS40:Slice	15.40	331.80	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=215	40	STND PLUS	SS40:Slice	11.41	382.96	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=200	40	STND PLUS	SS40:Slice	10.62	356.22	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	130	40	STND PLUS	SS40:Slice	4.94	122.39	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	60	40	STND	SS40:Slice	33.61	537.71	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.17	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE FULL	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.19	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375 55.00 11 pass	0.4	5	40	120	SmartmA 50-500mA NI=14 Avg mA=210	40	STND PLUS	SS40	73.97	2330.01	30.00	Body 32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature
25.21	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40:Slice	38.46	1288.40	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.22	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	260	40	STND SEGM	SS40:Slice	5.86	58.59	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.24	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40	38.60	1100.15	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	39.37 0.984	0.35	0.625	40	120	SmartmA 50-500mA NI=22 Avg mA=155	40	STND PLUS	SS40	4.80	256.89	49.94	Body 32	Retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSB	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.33	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSB	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.35	SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	330	40	STND SSSEG	SS40:Slice	14.45	202.35	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
25.37	SnapShot Assist Coronary		Helical	LargeBody	55.0 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries with SnapShot Assist.
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries with SnapShot Assist.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	6.4 0.16	0.35	0.625	40	120	ECG 250-450mA Phase 70-80 Avg mA=450	0	STND SSSEG	NONE	66.11	892.54	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries with SnapShot Assist.
25.38	SnapShot Assist Function		Helical	LargeBody	55.0 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of cardiac function with SnapShot Assist.
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of cardiac function with SnapShot Assist.
			Cardiac SEG	CardiacLarge	6.4 0.16	0.35	0.625	40	120	450	0	STND SSSEG	NONE	85.74	1157.48	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of cardiac function with SnapShot Assist.

Abdomen

Table 6-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Slice	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Slice	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-500mA NI=18 Avg mA=235	40	STND PLUS	SS40:Slice	16.63	724.35	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.5	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	130	40	STND PLUS	SS40:Slice	9.20	400.68	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.6	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Slice	9.20	400.68	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.8	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	170	40	STND PLUS	SS40:Slice	13.12	571.48	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.9	AAA + DMPR 0.625mm SmartmA/SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=170	40	STND PLUS	SS40:Slice	13.12	571.48	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.12	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.13	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	225.56	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-500mA NI=15 Avg mA=180	40	STND PLUS	SS40:Slice	9.56	129.47	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	267.16	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.14	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.15	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.16	VHS Liver Perfusion		Helical	LargeBody	39.37 0.984	0.6	5	40	120	100	40	STND PLUS	SS40	5.31	87.86	13.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical-S/P	LargeBody	1.375 55.00 21 pass	0.4	5	40	120	180	40	STND PLUS	SS40	152.38	2057.17	12.00	Body 32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.17	VHS 4D CTA		Helical	LargeBody	39.37 0.984	0.6	5	40	120	100	40	STND PLUS	SS40	5.31	87.86	13.00	Body 32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	STND	NONE	118.40	59.20	0.00	Body 32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375 55.00 18 pass	0.4	5	40	120	SmartmA 50-500mA NI=13 Avg mA=210	40	STND PLUS	SS40	135.28	2637.99	18.00	Body 32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.19	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	210	40	STND PLUS	SS40:Slice	16.21	1678.61	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.21	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-440mA NI=22.1 Avg mA=210	40	STND PLUS	SS40:Slice	16.21	1678.61	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.22	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 6-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	110	40	STND	SS40:Slice	29.81	59.62	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-440mA NI=14 Avg mA=155	40	BONE PLUS	SS40:Slice	30.46	637.58	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-440mA NI=20 Avg mA=220	40	BONE PLUS IQE+	SS40:Slice	54.03	1131.09	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-440mA NI=18 Avg mA=110	40	BONE PLUS IQE	SS40:Slice	27.02	565.50	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-440mA NI=20 Avg mA=220	40	BONE PLUS IQE+	SS40:Slice	54.03	1131.09	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 6-8
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=200	40	BONE PLUS IQE	SS40:Slice	17.70	241.74	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	10.62	303.06	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 6-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	75	40	STND PLUS	SS40:Si ce	12.67	119.48	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	95	40	STND PLUS IQE+	SS40:Si ce	16.67	86.95	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Si ce	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	75	40	STND PLUS IQE+	SS40:Si ce	13.16	68.65	3.50	Body 32	Helical scan mode 0.625mm , Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Si ce	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 64 Slice System, 72kW with 0.35sec/ ASiR

Head

Table 7-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	0	STND	SS30:Slice	15.22	182.63	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	235	0	STND	SS30:Slice	21.17	254.02	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	125	0	STND	SS30:Slice	22.38	268.57	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 7-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	SS20:Slice	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	105	0	STND PLUS IQE+	SS30:Slice	17.69	139.95	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	100	0	STND PLUS IQE+	SS30:Slice	16.85	94.32	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 7-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	55	0	STND PLUS	SS30:Slice	1.30	13.07	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSSEG	SS30:Slice	9.16	100.78	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSB	SS30:Slice	9.16	100.78	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 35-85 Avg mA=195	0	STND SSB	SS30:Slice	9.83	108.08	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-200mA NI=18 Avg mA=140	30	STND PLUS	SS30:Slice	1.45	16.05	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	85	0	STND PLUS	SS30:Slice	2.02	20.20	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	SS30:Slice	9.83	108.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSB	SS30:Slice	9.83	108.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 35-85 Avg mA=210	0	STND SSB	SS30:Slice	10.57	116.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-220mA NI=18 Avg mA=155	30	STND PLUS	SS30:Slice	1.61	17.77	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	70	0	STND PLUS	SS30:Slice	2.49	25.86	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSSEG	SS30:Slice	10.63	116.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSB	SS30:Slice	10.63	116.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 35-85 Avg mA=225	0	STND SSB	SS30:Slice	11.37	125.03	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-240mA NI=18 Avg mA=170	30	STND PLUS	SS30:Slice	1.76	19.48	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	SS30:Slice	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	SS30:Slice	11.31	124.38	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSB	SS30:Slice	11.31	124.38	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 35-85 Avg mA=240	0	STND SSB	SS30:Slice	12.11	133.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-280mA NI=18 Avg mA=175	30	STND PLUS	SS30:Slice	1.81	20.06	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	70	0	STND PLUS	SS30:Slice	2.49	28.66	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	12.43	136.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	12.43	136.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	13.35	146.86	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-300mA NI=18 Avg mA=210	30	STND PLUS	SS30:Slice	2.18	24.07	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	75	0	STND PLUS	SS30:Slice	2.67	30.70	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSSEG	SS30:Slice	13.11	144.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSB	SS30:Slice	13.11	144.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 35-85 Avg mA=280	0	STND SSB	SS30:Slice	14.10	155.12	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 100-340mA NI=18 Avg mA=240	30	STND PLUS	SS30:Slice	2.49	27.51	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	75	0	STND PLUS	SS30:Slice	2.85	42.10	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSSEG	SS30:Slice	21.67	238.39	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSB	SS30:Slice	21.67	238.39	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 35-85 Avg mA=245	0	STND SSB	SS30:Slice	23.39	257.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-360mA NI=18 Avg mA=190	30	STND PLUS	SS30:Slice	4.68	51.75	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	85	0	STND PLUS	SS30:Slice	3.23	54.18	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	23.60	259.64	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	23.60	259.64	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	25.36	278.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-400mA NI=16 Avg mA=280	30	STND PLUS	SS30:Slice	6.90	76.28	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	90	0	STND PLUS	SS30:Slice	3.42	67.63	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSSEG	SS30:Sli ce	27.01	297.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Sli ce	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSB	SS30:Sli ce	27.01	297.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Sli ce	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 35-85 Avg mA=295	0	STND SSB	SS30:Sli ce	28.59	314.46	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBo dy	39.37 0.984	0.5	0.625	40	100	SmartmA 100-440mA NI=16 Avg mA=280	30	STND PLUS	SS30:Sli ce	6.90	76.28	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 7-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	SS30:Sli ce	1.66	17.26	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	SS30:Sli ce	2.61	30.05	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	125	0	STND PLUS	SS30:Sli ce	2.96	37.48	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.52	34.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	95	0	STND PLUS	SS30:Sli ce	2.82	38.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	LargeBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.88	44.70	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	105	0	STND PLUS	SS30:Sli ce	3.33	65.78	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	SS30:Sli ce	3.80	75.17	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	140	0	STND PLUS	SS30:Sli ce	4.43	109.87	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 7-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND FULL	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND FULL	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND FULL	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 64 Slice System, 55kW with 0.35sec/ASiR

Head

Table 8-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	120	40	STND	SS40:Slice	42.97	257.83	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	60	40	STND	SS40:Slice	21.49	171.88	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	240	40	STND	SS40:Slice	43.24	605.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	120	40	STND PLUS IQE+	SS40:Slice	24.27	235.88	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=240	40	STND PLUS IQE+	SS40:Slice	48.83	915.01	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.7	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.8	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.9	VHS Head 4D CTA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature.
			Helical-S	Head	0.984 39.37 18 pass	0.5	5	40	120	145	40	STND PLUS	SS40	252.12	3151.55	11.00	Head 16	CT angiography of the head using Volume Helical scan mode for evaluation of cerebral vasculature.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.11	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.12	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.13	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 8-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	90	40	STND PLUS IQE+	SS40:Slice	30.33	277.68	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	22.24	175.90	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	100	40	STND PLUS IQE+	SS40:Slice	47.63	266.36	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 8-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	185	40	STND	SS40:Slice	21.63	302.85	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	145	40	STND	SS40:Slice	11.82	94.54	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-400mA NI=12.6 Avg mA=185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 50-400mA NI=12.6 Avg mA=185	40	BONE PLUS IQE+	SS40:Slice	21.86	315.57	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=115	40	STND PLUS IQE+	SS40:Slice	13.59	277.70	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA, Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=185	40	STND PLUS IQE+	SS40:Slice	11.45	235.24	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=115	40	STND PLUS IQE+	SS40:Slice	13.59	277.70	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=185	40	STND PLUS IQE+	SS40:Slice	11.45	235.24	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.12	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	20.62 0.516	0.6	0.625	40	120	SmartmA 80-400mA NI=11 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	13.29	271.71	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.14	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.8	2.5	40	120	SmartmA 80-400mA NI=9.1 Avg mA=95	40	STND PLUS	SS40:Slice	5.88	109.04	15.00	Body 32	Neck 2.5mm and SmartmA. Evaluation of soft tissue structures for abnormality
23.15	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	39.37 0.984	0.8	0.625	40	120	SmartmA 80-400mA NI=12.6 Avg mA=150	40	STND PLUS IQE+	SS40:Slice	9.29	172.17	15.00	Body 32	Neck with DMPR and SmartmA. Evaluation for soft tissue abnormalities

Upper Extremity

Table 8-4
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 2.5mm + DMPR	DMPR	Helical	LargeBo dy	20.62 0.516	1	2.5 [1.25]	40	140	110	40	STND PLUS	SS40:Sli ce	27.02	366.25	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 2.5mm + DMPR/ SmartmA	DMPR	Helical	LargeBo dy	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=110	40	STND PLUS	SS40:Sli ce	27.02	366.25	10.00	Body 32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 2.5mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBo dy	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=110	40	STND PLUS	SS40:Sli ce	27.02	366.25	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 8-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	117.68	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	55.00 1.375	0.6	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	117.68	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-400mA NI=12 Avg mA=125	40	CHEST PLUS	SS40:Slice	5.08	114.44	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	230	40	BONE+	SS40:Slice	3.10	74.28	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=125	40	STND PLUS	SS40:Slice	4.75	141.43	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Slice	6.90	159.07	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	61.25 1.531	0.8	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=155	40	STND PLUS	SS40:Slice	7.05	355.64	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	215	40	STND FULL	SS40:Slice	15.22	327.89	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	SmartmA 50-400mA NI=21.45 Avg mA=215	40	STND FULL	SS40:Slice	15.22	327.89	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=215	40	STND PLUS	SS40:Slice	11.41	382.96	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=200	40	STND PLUS	SS40:Slice	10.62	356.22	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	130	40	STND PLUS	SS40:Slice	4.94	122.39	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	60	40	STND	SS40:Slice	33.61	537.71	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.17	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE FULL	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.19	VHS Chest 4D CTA SmartPrep		Helical-S	LargeBody	1.375 55.00 11 pass	0.5	5	40	120	SmartmA 50-400mA NI=14 Avg mA=170	40	STND PLUS	SS40	69.49	2188.81	30.00	Body 32	Volume Helical scan mode with SmartPrep for time resolved CT angiography of chest vasculature
25.21	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40:Slice	38.46	1288.40	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.22	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	260	0	STND SEGM	SS40:Slice	5.86	58.59	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.24	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40	38.60	1100.15	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	39.37 0.984	0.35	0.625	40	120	SmartmA 50-400mA NI=22 Avg mA=155	40	STND PLUS	SS40	4.80	256.89	49.94	Body 32	Retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
25.26	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.27	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SSB	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSB	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.28	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	6.40 0.16	0.35	0.625	40	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSB	SS40:Slice	39.66	535.39	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.30	SnapShot Pulse		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Prospectively gated Cardiac Cine scan for evaluation of coronary arteries Intended for stable heart rates 30-65 bpm
			Cine	CardiacLarge	56i	0.35	0.625 [35]	40	120	330	0	STND SSSEG	SS40:Slice	14.45	202.35	13.94	Body 32	Prospectively gated Cardiac Cine scan for evaluation of coronary arteries. Intended for stable heart rates 30-65 bpm

Abdomen

Table 8-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	120	SmartmA 50-400mA NI=18 Avg mA=235	40	STND PLUS	SS40:Sli ce	16.63	724.35	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.5	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.6	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.8	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	125	40	STND PLUS	SS40:Sli ce	12.87	560.19	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.9	AAA + DMPR 0.625mm SmartmA/SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.8	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=125	40	STND PLUS	SS40:Sli ce	12.87	560.19	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Sli ce	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.12	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.13	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	225.56	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-400mA NI=15 Avg mA=180	40	STND PLUS	SS40:Slice	9.56	129.47	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	267.16	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.14	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.15	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.16	VHS Liver Perfusion		Helical	LargeBody	39.37 0.984	0.6	5	40	120	100	40	STND PLUS	SS40	5.31	87.86	13.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical-S/P	LargeBody	1.375 55.00 21 pass	0.5	5	40	120	145	40	STND PLUS	SS40	133.10	1796.84	12.00	Body 32	Volume Helical scan mode for evaluation of liver tumor perfusion over time
26.17	VHS 4D CTA		Helical	LargeBody	39.37 0.984	0.6	5	40	120	100	40	STND PLUS	SS40	5.31	87.86	13.00	Body 32	Localizer scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
		MIROI	Axial	LargeBody	1i	1	5 [0]	5	120	80	0	STND	NONE	118.40	59.20	0.00	Body 32	Timing bolus scan for Volume Helical Shuttle scan mode for evaluation of abdominal vasculature
			Helical-S	LargeBody	1.375 55.00 18 pass	0.5	5	40	120	SmartmA 50-400mA NI=13 Avg mA=170	40	STND PLUS	SS40	122.73	2393.26	18.00	Body 32	Volume Helical Shuttle scan mode for time resolved CT angiography of abdominal vasculature
26.19	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	205	40	STND PLUS	SS40:Slice	15.82	1638.65	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.21	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	0.625	40	140	SmartmA 50-340mA NI=22.1 Avg mA=205	40	STND PLUS	SS40:Slice	15.82	1638.65	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.22	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 8-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	110	40	STND	SS40:Slice	29.81	59.62	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-340mA NI=14 Avg mA=155	40	BONE PLUS	SS40:Slice	30.46	637.58	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-340mA NI=20 Avg mA=205	40	BONE PLUS IQE+	SS40:Slice	50.35	1053.97	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-340mA NI=18 Avg mA=110	40	BONE PLUS IQE	SS40:Slice	27.02	565.50	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	0.625	40	140	SmartmA 50-340mA NI=20 Avg mA=205	40	BONE PLUS IQE+	SS40:Slice	50.35	1053.97	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 8-8

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=200	40	BONE PLUS IQE	SS40:Slice	17.70	241.74	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	10.62	303.06	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 8-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	75	40	STND PLUS	SS40:Slice	12.67	119.48	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	95	40	STND PLUS IQE+	SS40:Slice	16.67	86.95	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	75	40	STND PLUS IQE+	SS40:Slice	13.16	68.65	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 64 Slice System, 55kW with 0.35sec/ ASiR

Head

Table 9-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	0	STND	SS30:Slice	15.22	182.63	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	235	0	STND	SS30:Slice	21.17	254.02	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	125	0	STND	SS30:Slice	22.38	268.57	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 9-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	SS20:Slice	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	105	0	STND PLUS IQE+	SS30:Slice	17.69	139.95	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	100	0	STND PLUS IQE+	SS30:Slice	16.85	94.32	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 9-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	55	0	STND PLUS	SS30:Slice	1.30	13.07	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSSEG	SS30:Slice	9.16	100.78	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSB	SS30:Slice	9.16	100.78	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-195mA Phase 35-85 Avg mA=195	0	STND SSB	SS30:Slice	9.83	108.08	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-200mA NI=18 Avg mA=140	30	STND PLUS	SS30:Slice	1.45	16.05	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	85	0	STND PLUS	SS30:Slice	2.02	20.20	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	SS30:Slice	9.83	108.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSB	SS30:Slice	9.83	108.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 55-210mA Phase 35-85 Avg mA=210	0	STND SSB	SS30:Slice	10.57	116.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-220mA NI=18 Avg mA=155	30	STND PLUS	SS30:Slice	1.61	17.77	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	70	0	STND PLUS	SS30:Slice	2.49	25.86	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSSEG	SS30:Slice	10.63	116.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSB	SS30:Slice	10.63	116.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-225mA Phase 35-85 Avg mA=225	0	STND SSB	SS30:Slice	11.37	125.03	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-240mA NI=18 Avg mA=170	30	STND PLUS	SS30:Slice	1.76	19.48	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	SS30:Slice	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	SS30:Slice	11.31	124.38	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSB	SS30:Slice	11.31	124.38	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-240mA Phase 35-85 Avg mA=240	0	STND SSB	SS30:Slice	12.11	133.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-280mA NI=18 Avg mA=175	30	STND PLUS	SS30:Slice	1.81	20.06	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	70	0	STND PLUS	SS30:Slice	2.49	28.66	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	12.43	136.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	12.43	136.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	13.35	146.86	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 50-300mA NI=18 Avg mA=210	30	STND PLUS	SS30:Slice	2.18	24.07	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	75	0	STND PLUS	SS30:Slice	2.67	30.70	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSSEG	SS30:Slice	13.11	144.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSB	SS30:Slice	13.11	144.17	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	80	ECG 70-280mA Phase 35-85 Avg mA=280	0	STND SSB	SS30:Slice	14.10	155.12	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	39.37 0.984	0.4	0.625	40	80	SmartmA 100-340mA NI=18 Avg mA=240	30	STND PLUS	SS30:Slice	2.49	27.51	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	75	0	STND PLUS	SS30:Slice	2.85	42.10	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSSEG	SS30:Slice	21.67	238.39	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSB	SS30:Slice	21.67	238.39	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 55-245mA Phase 35-85 Avg mA=245	0	STND SSB	SS30:Slice	23.39	257.25	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-360mA NI=18 Avg mA=190	30	STND PLUS	SS30:Slice	4.68	51.75	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	85	0	STND PLUS	SS30:Slice	3.23	54.18	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	23.60	259.64	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	23.60	259.64	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	6.40 0.16	0.35	0.625	40	100	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	25.36	278.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-400mA NI=16 Avg mA=280	30	STND PLUS	SS30:Slice	6.90	76.28	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	90	0	STND PLUS	SS30:Slice	3.42	67.63	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSEG	SS30:Slice	27.01	297.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSB	SS30:Slice	27.01	297.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	6.40 0.16	0.35	0.625	40	100	ECG 120-295mA Phase 35-85 Avg mA=295	0	STND SSB	SS30:Slice	28.59	314.46	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBody	39.37 0.984	0.5	0.625	40	100	SmartmA 100-440mA NI=16 Avg mA=280	30	STND PLUS	SS30:Slice	6.90	76.28	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 9-4
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	SS30:Slice	1.66	17.26	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	SS30:Slice	2.61	30.05	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	125	0	STND PLUS	SS30:Slice	2.96	37.48	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Slice	2.52	34.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	95	0	STND PLUS	SS30:Slice	2.82	38.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	LargeBody	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Slice	2.88	44.70	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	105	0	STND PLUS	SS30:Slice	3.33	65.78	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	SS30:Slice	3.80	75.17	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	140	0	STND PLUS	SS30:Slice	4.43	109.87	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 9-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND FULL	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND FULL	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND FULL	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 32 Slice System, 72kW with 0.35sec/ASiR-V

Head

Table 10-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	100	50	STND	AR50	35.81	214.86	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	50	50	STND	AR50	17.90	143.24	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	200	50	STND	AR50	35.81	501.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	STND PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=200	50	STND PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.6	Hi Res Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	DETL PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.7	Hi Res CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=200	50	DETL PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.9	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.10	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.14	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.15	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 10-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	75	50	STND PLUS IQE+	AR50	25.28	231.40	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	90	50	STND PLUS IQE+	AR40	18.20	143.92	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	85	50	STND PLUS IQE+	AR40	40.49	226.41	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 10-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	150	50	STND	AR70	17.41	243.78	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	120	50	STND	AR70	9.78	78.24	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-500mA NI=12.6 Avg mA=160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-500mA NI=12.6 Avg mA=160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=205	50	STND PLUS IQE+	AR40	10.42	196.77	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=205	50	STND PLUS IQE+	AR40	10.42	196.77	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Hi Res Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep	Prep	Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	DETL PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.12	Hi Res Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep	Prep	Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=205	50	DETL PLUS IQE+	AR40	10.42	196.77	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.14	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.15	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.16	Hi Res Carotid/ CoW 0.625mm SmartmA/ SmartPrep	Prep	Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=130	50	DETL PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.18	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.6	2.5	40	120	SmartmA 80-500mA NI=9.10 Avg mA=105	50	STND PLUS	AR50	4.87	90.42	15.00	Body 32	Neck 2.5mm and SmartmA, Evaluation of soft tissue structures for abnormality
23.19	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=12.60 Avg mA=165	50	STND PLUS IQE+	AR40	8.33	140.43	15.00	Body 32	Neck with DMPR and SmartmA, Evaluation for soft tissue abnormalities

Upper Extremity

Table 10-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR and Smart mA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 1.25mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=110	50	STND PLUS IQE	AR50	27.02	362.87	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 10-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	98.85	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	27.50 1.375	0.6	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=105	50	STND PLUS	AR40	4.27	96.13	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-500mA NI=12 Avg mA=105	50	CHEST PLUS	AR50	4.27	96.13	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	BONE+	AR50	2.54	60.92	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	118.80	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	5.84	134.60	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	30.62 1.531	0.8	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=125	50	STND PLUS	AR40	6.09	290.65	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	240	50	STND	AR50	12.74	274.59	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=21.45 Avg mA=240	50	STND	AR50	12.74	274.59	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=175	50	STND PLUS	AR40	10.10	321.84	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	STND PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Hi Res Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=175	50	DETL PLUS	AR40	10.10	321.84	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.5:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection.
25.17	Hi Res Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	DETL PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection.
25.19	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	110	50	STND PLUS	AR50	4.18	103.56	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	50	50	STND	AR50	28.01	448.09	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.20	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.24	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	39.03	1239.09	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.25	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	215	50	STND SEGM	AR50	4.84	48.45	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.27	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	39.10	1045.88	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Helical	LargeBody	19.37 0.969	0.35	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=125	50	STND PLUS	AR40	4.21	218.08	49.94	Body 32	
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSSEG	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.33	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	STND SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.35	Hi Res SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSSEG	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30- 74 bpm
25.36	Hi Res SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.37	Hi Res SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	50	DETL SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm

Abdomen

Table 10-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/ MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-500mA NI=18 Avg mA=190	50	STND PLUS	AR40	14.62	612.09	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.4	Hi Res Abdomen Pelvis +DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-500mA NI=18 Avg mA=190	50	DETL PLUS	AR40	14.62	612.09	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.6	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.7	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.9	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	135	50	STND PLUS	AR40	11.33	474.30	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.10	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=135	50	STND PLUS	AR40	11.33	474.30	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Hi Res AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	135	50	DETL PLUS	AR40	11.33	474.30	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.12	Hi Res AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=135	50	DETL PLUS	AR40	11.33	474.30	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.14	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.15	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.16	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	189.95	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-500mA NI=15 Avg mA=150	50	STND PLUS	AR50	7.96	107.89	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	222.64	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.17	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.18	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.23	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	170	50	STND PLUS	AR40	14.27	1453.34	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.25	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=170	50	STND PLUS	AR40	14.27	1453.34	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.26	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.27	Hi Res Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=170	50	DETL PLUS	AR40	14.27	1453.34	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.28	Hi Res Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=125	50	DETL PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.30	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 10-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	90	50	STND	AR60	24.39	48.78	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-440mA NI=14 Avg mA=130	50	BONE PLUS	AR50	25.54	534.75	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-440mA NI=20 Avg mA=185	50	BONE PLUS IQE+	AR50	47.54	914.33	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-440mA NI=18 Avg mA=90	50	BONE PLUS IQE	AR50	22.11	462.69	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-440mA NI=20 Avg mA=185	50	BONE PLUS IQE+	AR50	47.54	914.33	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 10-8

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=165	50	BONE PLUS IQE	AR50	14.60	199.44	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=125	50	STND PLUS	AR50	8.85	252.55	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 10-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	60	50	STND PLUS	AR50	10.14	95.59	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	80	50	STND PLUS IQE+	AR40	14.03	73.22	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	60	50	STND PLUS IQE+	AR40	10.53	54.92	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 32

Slice System, 72kW with 0.35sec/

ASiR-V

Head

Table 11-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	145	0	STND	AR40	12.98	155.77	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	200	0	STND	AR40	17.90	214.86	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	110	0	STND	AR40	19.70	236.34	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 11-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	AR20	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	90	0	STND PLUS IQE+	AR40	15.17	119.96	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	85	0	STND PLUS IQE+	AR40	14.32	80.17	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 11-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	50	0	STND PLUS	AR40	1.19	11.88	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSSEG	AR30	9.59	88.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSB	AR30	9.59	88.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 35-85 Avg mA=170	0	STND SSB	AR30	10.35	95.75	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-200mA NI=18 Avg mA=115	40	STND PLUS	AR30	1.30	12.13	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	AR40	1.66	16.64	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSSEG	AR30	10.12	93.60	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSB	AR30	10.12	93.60	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 35-85 Avg mA=180	0	STND SSB	AR30	10.94	101.24	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-220mA NI=18 Avg mA=130	40	STND PLUS	AR30	1.46	13.72	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	60	0	STND PLUS	AR40	2.13	22.16	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSSEG	AR30	10.76	99.57	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSB	AR30	10.76	99.57	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 35-85 Avg mA=190	0	STND SSB	AR30	11.59	107.21	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-240mA NI=18 Avg mA=140	40	STND PLUS	AR30	1.58	14.77	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	55	0	STND PLUS	AR40	1.96	22.52	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSSEG	AR30	11.57	106.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSB	AR30	11.57	106.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 35-85 Avg mA=205	0	STND SSB	AR30	12.49	115.51	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-280mA NI=18 Avg mA=145	40	STND PLUS	AR30	1.63	15.30	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	60	0	STND PLUS	AR40	2.13	24.56	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	12.90	119.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	12.90	119.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	13.98	129.33	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-300mA NI=18 Avg mA=175	40	STND PLUS	AR30	1.97	18.47	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	AR40	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	AR30	13.44	124.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSB	AR30	13.44	124.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 35-85 Avg mA=240	0	STND SSB	AR30	14.59	134.93	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 100-340mA NI=18 Avg mA=200	40	STND PLUS	AR30	2.25	21.10	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	65	0	STND PLUS	AR40	2.47	36.49	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	AR30	22.32	206.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSB	AR30	22.32	206.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 35-85 Avg mA=210	0	STND SSB	AR30	24.25	224.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.4	0.625	20	100	SmartmA 100-360mA NI=18 Avg mA=200	40	STND PLUS	AR30	4.29	40.16	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	70	0	STND PLUS	AR40	2.66	44.62	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	24.55	227.10	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	24.55	227.10	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	26.61	246.11	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-400mA NI=16 Avg mA=235	40	STND PLUS	AR30	6.30	58.95	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	80	0	STND PLUS	AR40	3.04	60.11	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSSEG	AR30	27.65	255.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSB	AR30	27.65	255.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 35-85 Avg mA=250	0	STND SSB	AR30	29.48	272.67	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-440mA NI=16 Avg mA=235	40	STND PLUS	AR30	6.30	58.95	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 11-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	60	0	STND PLUS	AR40	1.42	14.79	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	95	0	STND PLUS	AR40	2.25	25.95	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	AR40	2.61	32.99	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	75	0	STND PLUS	AR40	2.22	30.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.5	5	20	120	80	0	STND PLUS	AR40	2.37	32.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	LargeBody	27.50 1.375	0.5	5	20	120	70	0	STND PLUS	AR40	2.37	36.81	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5-31.5kg (49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	90	0	STND PLUS	AR40	2.85	56.38	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5-40.5kg (69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	100	0	STND PLUS	AR40	3.17	62.65	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5-55.0kg (89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	AR40	3.80	94.18	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 11-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 32 Slice System, 55kW with 0.35sec/ASiR-V

Head

Table 12-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	100	50	STND	AR50	35.81	214.86	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	50	50	STND	AR50	17.90	143.24	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	200	50	STND	AR50	35.81	501.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	STND PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=200	50	STND PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.6	Hi Res Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	100	50	DETL PLUS IQE+	AR40	20.22	196.56	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.7	Hi Res CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=200	50	DETL PLUS IQE+	AR40	40.44	757.12	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.9	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.10	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	90	50	STND PLUS IQE+	AR40	24.27	393.56	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.13	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.14	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.15	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	85	50	STND	AR50	30.44	426.13	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 12-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	75	50	STND PLUS IQE+	AR50	25.28	231.40	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	90	50	STND PLUS IQE+	AR40	18.20	143.92	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	85	50	STND PLUS IQE+	AR40	40.49	226.41	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	80	50	STND	AR60	40.49	161.96	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 12-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	150	50	STND	AR70	17.41	243.78	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	120	50	STND	AR70	9.78	78.24	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-400mA NI=12.6 Avg mA=160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-400mA NI=12.6 Avg mA=160	50	BONE PLUS IQE+	AR50	19.64	249.76	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=95	50	STND PLUS IQE+	AR40	11.66	218.25	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA, Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=150	50	STND PLUS IQE+	AR40	10.10	190.38	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=95	50	STND PLUS IQE+	AR40	11.66	218.25	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=150	50	STND PLUS IQE+	AR40	10.10	190.38	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Hi Res Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=95	50	DETL PLUS IQE+	AR40	11.66	218.25	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.12	Hi Res Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=150	50	DETL PLUS IQE+	AR40	10.10	190.38	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.14	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.15	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=130	50	STND PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.16	Hi Res Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=130	50	DETL PLUS IQE+	AR40	11.97	224.01	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
23.18	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.8	2.5	40	120	SmartmA 80-400mA NI=9.1 Avg mA=80	50	STND PLUS	AR50	4.95	91.82	15.00	Body 32	Neck 2.5mm and SmartmA, Evaluation of soft tissue structures for abnormality
23.19	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=12.6 Avg mA=120	50	STND PLUS IQE+	AR40	8.08	136.15	15.00	Body 32	Neck with DMPR and SmartmA, Evaluation for soft tissue abnormalities

Upper Extremity

Table 12-4
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 2.5mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 2.5mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR and Smart mA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 2.5mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=90	50	STND PLUS	AR50	22.11	299.66	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 12-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	98.85	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	27.50 1.375	0.6	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=105	50	STND PLUS	AR40	4.27	96.13	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-400mA NI=12 Avg mA=105	50	CHEST PLUS	AR50	4.27	96.13	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	190	50	BONE+	AR50	2.54	60.92	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=105	50	STND PLUS	AR50	3.99	118.80	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	5.84	134.60	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	30.62 1.531	0.8	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=125	50	STND PLUS	AR40	6.09	290.65	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	180	50	STND	AR50	12.74	274.48	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	SmartmA 50-400mA NI=21.45 Avg mA=180	50	STND	AR50	12.74	274.48	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=175	50	STND PLUS	AR40	10.10	321.84	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	STND PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Hi Res Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=175	50	DETL PLUS	AR40	10.10	321.84	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.5:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.17	Hi Res Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	DETL PLUS	AR50	8.76	293.88	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.19	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	110	50	STND PLUS	AR50	4.18	103.56	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	50	50	STND	AR50	28.01	448.09	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
25.20	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.24	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	39.03	1239.09	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.25	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	215	0	STND SEGM	AR50	4.84	48.45	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.27	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	39.10	1045.88	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	19.37 0.969	0.35	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=125	50	STND PLUS	AR40	4.21	218.08	49.94	Body 32	
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSSEG	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.33	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	STND SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm
25.35	Hi Res SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSSEG	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.36	Hi Res SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm
25.37	Hi Res SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 125-225mA Phase 70-80 Avg mA=225	0	DETL SSB	AR40	39.68	466.24	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm

Abdomen

Table 12-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/ MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-400mA NI=18 Avg mA=190	50	STND PLUS	AR40	14.62	612.09	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.4	Hi Res Abdomen Pelvis +DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-400mA NI=18 Avg mA=190	50	DETL PLUS	AR40	14.62	612.09	40.00	Body 32	Basic Abdomen/ Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.6	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.7	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=110	50	STND PLUS	AR50	7.79	339.04	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.9	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	100	50	STND PLUS	AR40	11.19	468.41	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.10	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=100	50	STND PLUS	AR40	11.19	468.41	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Hi Res AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	100	50	DETL PLUS	AR40	11.19	468.41	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.12	Hi Res AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=100	50	DETL PLUS	AR40	11.19	468.41	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.14	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.15	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.16	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	189.95	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-400mA NI=15 Avg mA=150	50	STND PLUS	AR50	7.96	107.89	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	222.64	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.17	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.18	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=80	50	STND PLUS	AR50	5.66	246.57	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	156.27	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	6.64	289.00	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASIR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
26.23	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	165	50	STND PLUS	AR40	13.85	1410.59	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.24	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.25	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=165	50	STND PLUS	AR40	13.85	1410.59	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.26	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=125	50	STND PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.27	Hi Res Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=165	50	DETL PLUS	AR40	13.85	1410.59	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.28	Hi Res Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=125	50	DETL PLUS	AR50	9.65	999.15	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.30	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	60	50	STND PLUS	AR50	3.19	138.72	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 12-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	80	50	STND	AR60	14.91	29.82	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	90	50	STND	AR60	24.39	48.78	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-340mA NI=14 Avg mA=130	50	BONE PLUS	AR50	25.54	534.75	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-240mA NI=18 Avg mA=90	50	BONE PLUS IQE	AR50	27.02	565.5	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-240mA NI=18 Avg mA=170	50	BONE PLUS IQE	AR50	27.02	565.5	17.50	Body 32	Helical scan mode 1.25mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 12-8

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=165	50	BONE PLUS IQE	AR50	14.60	199.44	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=125	50	STND PLUS	AR50	8.85	252.55	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 12-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	60	50	STND PLUS	AR50	10.14	95.59	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	80	50	STND PLUS IQE+	AR40	14.03	73.22	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	60	50	STND PLUS IQE+	AR40	10.53	54.92	3.50	Body 32	Helical scan mode 0.625mm , Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	AR30	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 32

Slice System, 55kW with 0.35sec/

ASiR-V

Head

Table 13-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	145	0	STND	AR40	12.98	155.77	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	200	0	STND	AR40	17.90	214.86	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	110	0	STND	AR40	19.70	236.34	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 13-2
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	AR20	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	90	0	STND PLUS IQE+	AR40	15.17	119.96	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	85	0	STND PLUS IQE+	AR40	14.32	80.17	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 13-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	50	0	STND PLUS	AR40	1.19	11.88	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSSEG	AR30	9.59	88.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 40-80 Avg mA=170	0	STND SSB	AR30	9.59	88.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	50	0	STND PLUS	AR40	0.56	5.43	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-170mA Phase 35-85 Avg mA=170	0	STND SSB	AR30	10.35	95.75	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-200mA NI=18 Avg mA=115	40	STND PLUS	AR30	1.30	12.13	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	AR40	1.66	16.64	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSSEG	AR30	10.12	93.60	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 40-80 Avg mA=180	0	STND SSB	AR30	10.12	93.60	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 50-180mA Phase 35-85 Avg mA=180	0	STND SSB	AR30	10.94	101.24	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-220mA NI=18 Avg mA=130	40	STND PLUS	AR30	1.46	13.72	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	60	0	STND PLUS	AR40	2.13	22.16	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSSEG	AR30	10.76	99.57	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 40-80 Avg mA=190	0	STND SSB	AR30	10.76	99.57	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-190mA Phase 35-85 Avg mA=190	0	STND SSB	AR30	11.59	107.21	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-240mA NI=18 Avg mA=140	40	STND PLUS	AR30	1.58	14.77	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	55	0	STND PLUS	AR40	1.96	22.52	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSSEG	AR30	11.57	106.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 40-80 Avg mA=205	0	STND SSB	AR30	11.57	106.99	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-205mA Phase 35-85 Avg mA=205	0	STND SSB	AR30	12.49	115.51	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-280mA NI=18 Avg mA=145	40	STND PLUS	AR30	1.63	15.30	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	60	0	STND PLUS	AR40	2.13	24.56	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	12.90	119.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	12.90	119.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	13.98	129.33	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-300mA NI=18 Avg mA=175	40	STND PLUS	AR30	1.97	18.47	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	AR40	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	AR30	13.44	124.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 40-80 Avg mA=240	0	STND SSB	AR30	13.44	124.35	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	70	0	STND PLUS	AR40	0.78	7.61	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 60-240mA Phase 35-85 Avg mA=240	0	STND SSB	AR30	14.59	134.93	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 100-340mA NI=18 Avg mA=200	40	STND PLUS	AR30	2.25	21.10	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	65	0	STND PLUS	AR40	2.47	36.49	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	AR30	22.32	206.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 40-80 Avg mA=210	0	STND SSB	AR30	22.32	206.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 50-210mA Phase 35-85 Avg mA=210	0	STND SSB	AR30	24.25	224.34	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy-cm)	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.4	0.625	20	100	SmartmA 100-360mA NI=18 Avg mA=200	40	STND PLUS	AR30	4.29	40.16	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	70	0	STND PLUS	AR40	2.66	44.62	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSSEG	AR30	24.55	227.10	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.12	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 40-80 Avg mA=230	0	STND SSB	AR30	24.55	227.10	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 60-230mA Phase 35-85 Avg mA=230	0	STND SSB	AR30	26.61	246.11	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-400mA NI=16 Avg mA=235	40	STND PLUS	AR30	6.30	58.95	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	80	0	STND PLUS	AR40	3.04	60.11	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSSEG	AR30	27.65	255.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 40-80 Avg mA=250	0	STND SSB	AR30	27.65	255.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	70	0	STND PLUS	AR40	1.48	14.48	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 100-250mA Phase 35-85 Avg mA=250	0	STND SSB	AR30	29.48	272.67	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-440mA NI=16 Avg mA=235	40	STND PLUS	AR30	6.30	58.95	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 13-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·c m)	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	60	0	STND PLUS	AR40	1.42	14.79	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	95	0	STND PLUS	AR40	2.25	25.95	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5- 11.5kg (20.9- 25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	AR40	2.61	32.99	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5- 14.5kg (25.4- 32.0lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	75	0	STND PLUS	AR40	2.22	30.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5- 18.5kg (32.0- 40.8lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	80	0	STND PLUS	AR40	2.37	32.04	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5- 22.5kg (40.8- 49.6lbs)		Helical	LargeBo dy	27.50 1.375	0.5	5	20	120	70	0	STND PLUS	AR40	2.37	36.81	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	90	0	STND PLUS	AR40	2.85	56.38	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	100	0	STND PLUS	AR40	3.17	62.65	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	AR40	3.80	94.18	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 13-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR-V (%)	CTDI vol (mGy)	DLP (mGy·cm)	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 32 Slice System, 72kW with 0.35sec/ASiR

Head

Table 14-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	120	40	STND	SS40:Slice	42.97	257.83	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	60	40	STND	SS40:Slice	21.49	171.88	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	240	40	STND	SS40:Slice	43.24	605.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	120	40	STND PLUS IQE+	SS40:Slice	24.27	235.88	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-500mA NI=8 Avg mA=240	40	STND PLUS IQE+	SS40:Slice	48.83	915.01	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.7	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.8	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.11	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.12	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.13	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 14-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	90	40	STND PLUS IQE+	SS40:Slice	30.33	277.68	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	22.24	175.90	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	100	40	STND PLUS IQE+	SS40:Slice	47.63	266.36	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 14-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	185	40	STND	SS40:Slice	21.63	302.85	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	145	40	STND	SS40:Slice	11.82	94.54	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-500mA NI=12.6 Avg mA=190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-500mA NI=12.6 Avg mA=190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=245	40	STND PLUS IQE+	SS40:Slice	12.46	235.16	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=245	40	STND PLUS IQE+	SS40:Slice	12.46	235.16	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.12	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-500mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.14	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.6	2.5	40	120	SmartmA 80-500mA NI=9.1 Avg mA=125	40	STND PLUS	SS40:Slice	5.80	107.64	15.00	Body 32	Neck 2.5mm and SmartmA. Evaluation of soft tissue structures for abnormality
23.15	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	19.37 0.969	0.6	0.625	20	120	SmartmA 80-500mA NI=12.6 Avg mA=200	40	STND PLUS IQE+	SS40:Slice	10.10	170.22	15.00	Body 32	Neck with DMPR and SmartmA. Evaluation for soft tissue abnormalities

Upper Extremity

Table 14-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 1.25mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR and SmartmA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 1.25mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	1.25 [0.625]	40	140	SmartmA 100-440mA NI=21.45 Avg mA=130	40	STND PLUS IQE	SS40:Slice	31.93	428.85	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 14-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=125	40	STND PLUS	SS40:Si ce	4.75	117.68	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	27.50 1.375	0.6	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=125	40	STND PLUS	SS40:Si ce	5.08	114.44	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-500mA NI=12 Avg mA=125	40	CHEST PLUS	SS40:Si ce	5.08	114.44	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	230	40	BONE+	SS40:Si ce	3.10	74.28	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-500mA NI=13 Avg mA=125	40	STND PLUS	SS40:Si ce	4.75	141.43	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Si ce	6.90	159.07	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	30.62 1.531	0.8	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=155	40	STND PLUS	SS40:Si ce	7.55	360.40	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolism	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	290	40	STND FULL	SS40:Slice	15.40	331.80	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolism SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=21.45 Avg mA=290	40	STND FULL	SS40:Slice	15.40	331.80	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=215	40	STND PLUS	SS40:Slice	12.50	398.57	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=200	40	STND PLUS	SS40:Slice	10.62	356.22	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	130	40	STND PLUS	SS40:Slice	4.94	122.39	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	60	40	STND	SS40:Slice	33.61	537.71	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.17	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE FULL	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.21	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40:Slice	46.85	1487.35	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.22	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	260	40	STND SEGM	SS40:Slice	5.86	58.59	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.24	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40	46.93	1255.42	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Helical	LargeBody	19.37 0.969	0.35	0.625	20	120	SmartmA 50-500mA NI=22 Avg mA=150	40	STND PLUS	SS40	5.05	261.70	49.94	Body 32	Retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
25.31	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSSEG	SS40:Slice	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.32	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSB	SS40:Slice	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.33	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	40	STND SSB	SS40:Silice	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm

Abdomen

Table 14-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/ MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-500mA NI=18 Avg mA=230	40	STND PLUS	SS40:Sli ce	17.83	746.78	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.5	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.6	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.8	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	165	40	STND PLUS	SS40:Sli ce	13.85	579.70	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.9	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=165	40	STND PLUS	SS40:Sli ce	13.85	579.70	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Sli ce	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.12	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.13	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	225.56	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-500mA NI=15 Avg mA=180	40	STND PLUS	SS40:Slice	9.56	129.47	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	267.16	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.14	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.15	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.19	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	205	40	STND PLUS	SS40:Slice	17.33	1765.82	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.21	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-440mA NI=22.1 Avg mA=205	40	STND PLUS	SS40:Slice	17.33	1765.82	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.22	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-440mA NI=21.45 Avg mA=150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vasacular structures of abdomen, femurs and lower extremities (Runoff)
26.24	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 14-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	110	40	STND	SS40:Slice	29.81	59.62	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and boney structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-440mA NI=14 Avg mA=155	40	BONE PLUS	SS40:Slice	30.46	637.58	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-440mA NI=20 Avg mA=225	40	BONE PLUS IQE+	SS40:Slice	57.82	1112.03	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-440mA NI=18 Avg mA=110	40	BONE PLUS IQE	SS40:Slice	27.02	565.50	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-440mA NI=20 Avg mA=225	40	BONE PLUS IQE+	SS40:Slice	57.82	1112.03	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 14-8

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-500mA NI=20 Avg mA=200	40	BONE PLUS IQE	SS40:Slice	17.70	241.74	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-500mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	10.62	303.06	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 14-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	75	40	STND PLUS	SS40:Slice	12.67	119.48	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	95	40	STND PLUS IQE+	SS40:Slice	16.67	86.95	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	75	40	STND PLUS IQE+	SS40:Slice	13.16	68.65	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 32

Slice System, 72kW with 0.35sec/ ASiR

Head

Table 15-1
[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	0	STND	SS30:Slice	15.22	182.63	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	235	0	STND	SS30:Slice	21.17	254.02	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	125	0	STND	SS30:Slice	22.38	268.57	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 15-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	SS20:Slice	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	105	0	STND PLUS IQE+	SS30:Slice	17.69	139.95	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	100	0	STND PLUS IQE+	SS30:Slice	16.85	94.32	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 15-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	55	0	STND PLUS	SS30:Slice	1.30	13.07	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSSEG	SS30:Slice	10.98	101.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSB	SS30:Slice	10.98	101.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 35-85 Avg mA=195	0	STND SSB	SS30:Slice	11.87	109.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-200mA NI=18 Avg mA=135	30	STND PLUS	SS30:Slice	1.52	14.25	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	85	0	STND PLUS	SS30:Slice	2.02	20.20	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	SS30:Slice	11.78	108.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSB	SS30:Slice	11.78	108.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 35-85 Avg mA=210	0	STND SSB	SS30:Slice	12.76	118.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-220mA NI=18 Avg mA=150	30	STND PLUS	SS30:Slice	1.69	15.83	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	70	0	STND PLUS	SS30:Slice	2.49	25.86	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSSEG	SS30:Slice	12.75	117.92	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSB	SS30:Slice	12.75	117.92	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 35-85 Avg mA=225	0	STND SSB	SS30:Slice	13.73	127.02	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-240mA NI=18 Avg mA=165	30	STND PLUS	SS30:Slice	1.86	17.41	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	SS30:Slice	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	SS30:Slice	13.55	125.37	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSB	SS30:Slice	13.55	125.37	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 35-85 Avg mA=240	0	STND SSB	SS30:Slice	14.63	135.36	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-280mA NI=18 Avg mA=170	30	STND PLUS	SS30:Slice	1.91	17.94	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	70	0	STND PLUS	SS30:Slice	2.49	28.66	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	14.89	137.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	14.89	137.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	16.13	149.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-300mA NI=18 Avg mA=205	30	STND PLUS	SS30:Slice	2.31	21.63	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	75	0	STND PLUS	SS30:Slice	2.67	30.70	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSSEG	SS30:Slice	15.70	145.20	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSB	SS30:Slice	15.70	145.20	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 35-85 Avg mA=280	0	STND SSB	SS30:Slice	17.03	157.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 100-340mA NI=18 Avg mA=235	30	STND PLUS	SS30:Slice	2.65	24.80	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	75	0	STND PLUS	SS30:Slice	2.85	42.10	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSSEG	SS30:Slice	26.13	241.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSB	SS30:Slice	26.13	241.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 35-85 Avg mA=245	0	STND SSB	SS30:Slice	28.44	263.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-360mA NI=18 Avg mA=185	30	STND PLUS	SS30:Slice	4.96	46.41	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	85	0	STND PLUS	SS30:Slice	3.23	54.18	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	28.48	263.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	28.48	263.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	30.86	285.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-400mA NI=16 Avg mA=275	30	STND PLUS	SS30:Slice	7.42	69.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	90	0	STND PLUS	SS30:Slice	3.42	67.63	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSSEG	SS30:Slice	32.69	302.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSB	SS30:Slice	32.69	302.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	2.80 0.14	0.35	0.625	40	100	ECG 120-295mA Phase 35-85 Avg mA=295	0	STND SSB	SS30:Slice	34.82	322.13	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-440mA NI=16 Avg mA=275	30	STND PLUS	SS30:Slice	7.42	69.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 15-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	SS30:Sli ce	1.66	17.26	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	SS30:Sli ce	2.61	30.05	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5-11.5kg (20.9-25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	125	0	STND PLUS	SS30:Sli ce	2.96	37.48	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.52	34.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	95	0	STND PLUS	SS30:Sli ce	2.82	38.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5-22.5kg (40.8-49.6lbs)		Helical	LargeBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.88	44.70	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	105	0	STND PLUS	SS30:Sli ce	3.33	65.78	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	SS30:Sli ce	3.80	75.17	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	140	0	STND PLUS	SS30:Sli ce	4.43	109.87	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 15-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND FULL	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND FULL	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND FULL	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Reference Adult Protocols 32 Slice System, 55kW with 0.35sec/ASiR

Head

Table 16-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
21.1	Routine Head		Axial	Head	4i	2	5	20	120	120	40	STND	SS40:Slice	42.97	257.83	5.50	Head 16	Routine head protocol for evaluation of the brain for abnormalities
			Axial	Head	2i	2	10	20	120	60	40	STND	SS40:Slice	21.49	171.88	7.00	Head 16	Routine head protocol for evaluation of the brain for abnormalities
21.2	Trauma Head		Axial	Head	4i	1	5	20	120	240	40	STND	SS40:Slice	43.24	605.33	13.50	Head 16	Emergency head protocol for evaluation of the brain and cranium for abnormalities.
21.4	Circle of Willis	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	118.25	59.12	0	Head 16	Timing bolus for CT angiography of the head for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	120	40	STND PLUS IQE+	SS40:Slice	24.27	235.88	8.00	Head 16	CT angiography of the head for evaluation of carotid and cerebral vasculature
21.5	CoW/ Carotid 0.625mm SmartmA	MIROI	Axial	Head	1i	1	5 [0]	5	120	50	0	STND	NONE	106.42	53.21	0	Head 16	Timing Bolus for CT angiography of the neck and head using SmartmA with ODM for evaluation of carotid and cerebral vasculature
			Helical	Head	10.62 0.531	0.6	0.625	20	120	SmartmA 50-400mA NI=8 Avg mA=240	40	STND PLUS IQE+	SS40:Slice	48.83	915.01	17.00	Head 16	CT angiography of the neck and head using SmartmA with ODM for evaluation of cerebral vasculature
21.7	Helical Head	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode for evaluation of the brain for cerebral abnormality
21.8	Helical Head MAR	DMPR	Helical	Head	10.62 0.531	0.8	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	29.66	481.02	14.50	Head 16	Helical scan mode with SmartMAR for evaluation of the brain for cerebral abnormality
21.11	CT Perfusion 350-370 Strength Contrast		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Cine	Head	8i	1	5 [0]	40	80	180	0	STND	NONE	478.96	1915.83	3.50	Head 16	CT Perfusion using Cine scan mode for evaluation of cerebral perfusion over time

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.12	CT Perfusion 350-370 Strength Contrast (Axial)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial	Head	8i	1	5 [0]	40	80	200	0	STND	NONE	260.12	1040.47	3.50	Head 16	CT Perfusion using Axial scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement
21.13	CT Perfusion 350-370 Strength Contrast (Axial Shuttle Mode)		Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Non-Enhance Brain, Evaluation for hemorrhage or infarction
		Perfusion	Axial-S	Head	8i 17 pass	0.5	5	40	80	400	0	STND	NONE	201.00	1607.99	7.50	Head 16	CT Perfusion using Axial Shuttle scan mode for evaluation of cerebral perfusion over time
			Axial	Head	4i	2	5	20	120	100	40	STND	SS40:Slice	35.81	501.33	13.50	Head 16	Post Contrast Brain, Evaluation for residual tissue enhancement

Orbit

Table 16-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
22.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	1	0.625	20	120	90	40	STND PLUS IQE+	SS40:Slice	30.33	277.68	7.44	Head 16	Routine Helical Sinus protocol, Evaluation for any abnormality
22.3	Orbits Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.6	0.625	20	120	110	40	STND PLUS IQE+	SS40:Slice	22.24	175.90	6.19	Head 16	Routine Helical Orbit protocol, Evaluation for abnormality soft tissues and bone structures of the face
22.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	1	0.625	20	140	100	40	STND PLUS IQE+	SS40:Slice	47.63	266.36	3.88	Head 16	Helical scan mode, Evaluation for inner ear abnormalities in Axial plane
22.7	Axial and Coronal IAC		Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Axial plane
			Axial	Head	16i	2	1.25	20	140	95	40	STND	SS40:Slice	48.08	192.32	3.88	Head 16	Axial scan mode, Evaluation for inner ear abnormalities in Coronal Plane

Neck

Table 16-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
23.1	C-Spine C5-C7 Axial		Axial	SmallBody	8i	1	2.5	20	140	185	40	STND	SS40:Slice	21.63	302.85	13.75	Body 32	C-Spine C5-C7 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.2	C-Spine C1-C4 Axial		Axial	SmallBody	8i	1	2.5	20	120	145	40	STND	SS40:Slice	11.82	94.54	7.75	Body 32	C-Spine C1-C4 area Axial mode, Evaluation for soft tissue, disc and bone abnormalities
23.3	C-Spine Helical + DMPR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR, Evaluation for soft tissue, disc and bone abnormalities
23.4	C-Spine Helical + DMPR/ SmartmA	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-400mA NI=12.6 Avg mA=190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR and SmartmA, Evaluation for soft tissue, disc and bone abnormalities
23.5	C-Spine Helical + DMPR/ SmartmA/ MAR	DMPR	Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 50-400mA NI=12.6 Avg mA=190	40	BONE PLUS IQE+	SS40:Slice	23.32	296.59	11.00	Body 32	C-Spine Helical mode with DMPR, SmartmA and SmartMAR, Evaluation for soft tissue, disc and bone abnormalities
23.7	Carotid 0.625mm 0.5:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	97.04	48.52	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.5:1) with SmartmA Evaluation of carotid vasculature
			Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=115	40	STND PLUS IQE+	SS40:Slice	14.11	264.19	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA, Evaluation of carotid vasculature
23.8	Carotid 0.625mm 0.9:1 SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	5 [0]	5	120	60	0	STND	NONE	64.69	32.35	0	Body 32	Timing bolus for Carotid CTA (Pitch 0.9:1) with SmartmA Evaluation of carotid vasculature

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=185	40	STND PLUS IQE+	SS40:Slice	12.45	234.80	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA. Evaluation of carotid vasculature
23.9	Carotid 0.625mm 0.5:1 SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=115	40	STND PLUS IQE+	SS40:Slice	14.11	264.19	17.00	Body 32	Carotid CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.10	Carotid 0.625mm 0.9:1 SmartmA/ SmartPrep		Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=185	40	STND PLUS IQE+	SS40:Slice	12.45	234.80	17.00	Body 32	Carotid CTA (Pitch 0.9:1) with SmartmA and SmartPrep. Evaluation of carotid vasculature
23.11	Carotid/ CoW 0.625mm SmartmA/ Test Bolus	MIROI	Axial	SmallBody	1i	1	10 [0]	10	120	60	0	STND	NONE	66.78	66.78	0	Body 32	Timing bolus for Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
			Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA. Evaluation of carotid and cerebral vasculature
23.12	Carotid/ CoW 0.625mm SmartmA/ SmartPrep		Helical	SmallBody	10.62 0.531	0.6	0.625	20	120	SmartmA 80-400mA NI=11 Avg mA=155	40	STND PLUS IQE+	SS40:Slice	14.27	267.09	17.00	Body 32	Carotid and brain CTA (Pitch 0.5:1) with SmartmA and SmartPrep. Evaluation of carotid and cerebral vasculature
23.14	Soft Tissue Neck 2.5mm SmartmA		Helical	SmallBody	39.37 0.984	0.8	2.5	40	120	SmartmA 80-400mA NI=9.1 Avg mA=95	40	STND PLUS	SS40:Slice	5.88	109.04	15.00	Body 32	Neck 2.5mm and SmartmA. Evaluation of soft tissue structures for abnormality
23.15	Soft Tissue Neck + DMPR/ SmartmA	DMPR	Helical	SmallBody	19.37 0.969	0.8	0.625	20	120	SmartmA 80-400mA NI=12.6 Avg mA=145	40	STND PLUS IQE+	SS40:Slice	9.76	164.51	15.00	Body 32	Neck with DMPR and SmartmA. Evaluation for soft tissue abnormalities

Upper Extremity

Table 16-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
24.1	Shoulder 2.5mm + DMPR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	110	40	STND PLUS	SS40:Slice	27.02	366.25	10.00	Body 32	Shoulder with DMPR, Evaluation of soft tissue and bone of shoulder area
24.2	Shoulder 2.5mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=110	40	STND PLUS	SS40:Slice	27.02	366.25	10.00	Body 32	Shoulder with DMPR and Smart mA, Evaluation of soft tissue and bone of shoulder area
24.3	Shoulder 2.5mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	20.62 0.516	1	2.5 [1.25]	40	140	SmartmA 50-340mA NI=21.45 Avg mA=110	40	STND PLUS	SS40:Slice	27.02	366.25	10.00	Body 32	Shoulder with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bone of shoulder area

Chest

Table 16-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.1	Routine Chest 5mm SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=125	40	STND PLUS	SS40:Sli ce	4.75	117.68	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lungs fields
25.2	Routine Chest + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	27.50 1.375	0.6	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=125	40	STND PLUS	SS40:Sli ce	5.08	114.44	20.00	Body 32	Chest with SmartmA, DMPR and SmartPrep, Evaluation of mediastinum and lung fields
25.4	Chest with HiRes SmartmA		Helical	LargeBody	27.50 1.375	0.6	5	20	120	SmartmA 50-400mA NI=12 Avg mA=125	40	CHEST PLUS	SS40:Sli ce	5.08	114.44	20.00	Body 32	Chest with Smart mA, Evaluation of mediastinum and lung fields with higher resolution images of the lungs
25.5	Hi Res Chest Axial		Axial	LargeBody	1i	0.5	1.25 [10]	1.25	120	230	40	BONE+	SS40:Sli ce	3.10	74.28	23.00	Body 32	Axial scan mode, Evaluation of lung fields in high resolution mode
25.7	Chest Abd Pelvis SmartmA		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 50-400mA NI=13 Avg mA=125	40	STND PLUS	SS40:Sli ce	4.75	141.43	25.00	Body 32	Multi group scan for Chest, Abdomen, Pelvis (0.6sec) with SmartmA and ODM, Evaluation of chest and abdominal anatomy for abnormalities
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	6.90	159.07	19.50	Body 32	
25.8	Chest Abd Pelvis + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	30.62 1.531	0.8	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=155	40	STND PLUS	SS40:Sli ce	7.55	360.40	45.00	Body 32	Helical scan mode (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation of chest and abdominal pelvic region for abnormalities

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.10	Pulmonary Embolis	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	215	40	STND FULL	SS40:Slice	15.22	327.89	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1, Evaluation of Chest for pulmonary embolism
25.11	Pulmonary Embolis SmartmA	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	42.44	42.44	0	Body 32	Timing Bolus for Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
			Helical	LargeBody	39.37 0.984	0.8	1.25	40	120	SmartmA 50-400mA NI=21.45 Avg mA=215	40	STND FULL	SS40:Slice	15.22	327.89	18.00	Body 32	Helical scan mode 1.25mm thickness and helical pitch of 0.9:1 with SmartmA, Evaluation of Chest for pulmonary embolism
25.13	Aortic Dissection 0.625mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.6	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=215	40	STND PLUS	SS40:Slice	12.50	398.57	30.00	Body 32	Helical scan mode 0.625mm thickness and helical pitch of 0.9:1 with SmartmA, SmartPrep and DMPR, Evaluation for aortic dissection
25.14	Aortic Dissection 1.25mm + DMPR/ SmartmA/ SmartPrep	DMPR	Helical	LargeBody	39.37 0.984	0.6	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=200	40	STND PLUS	SS40:Slice	10.62	356.22	30.00	Body 32	Helical scan of 1.25mm thickness and helical pitch of 0.9 with SmartmA, SmartPrep and DMPR for evaluation aortic dissection
25.16	Advantage 4D		Helical	LargeBody	55.00 1.375	0.6	5	40	120	130	40	STND PLUS	SS40:Slice	4.94	122.39	20.00	Body 32	Localization scan for Advantage 4D series
		Advantage 4D	Cine	LargeBody	8i	1	2.5	20	120	60	40	STND	SS40:Slice	33.61	537.71	15.75	Body 32	Cine scan mode for acquisition data for Advantage 4D for retrospective gating
25.17	Lung Assessment	Advanced Lung Analysis	Helical	LargeBody	19.37 0.969	0.5	1.25	20	120	120	0	BONE FULL	NONE	5.77	126.21	20.00	Body 32	Helical scan mode for acquisition of data for Advanced Lung Analysis (ALA) software for lung nodule assessment.
25.21	Gated Chest Cardiac Helical		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for Gated Chest scan for evaluation of thoracic vasculature
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for Gated Chest scan for evaluation of thoracic vascular

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40:Slice	46.85	1487.35	30.00	Body 32	Gated Chest scan for evaluation of thoracic vasculature
25.22	SmartScore - Gated	SmartScore	Cine	LargeBody	8i	0.35	2.5	20	120	260	0	STND SEGM	SS40:Slice	5.86	58.59	9.75	Body 32	Cine gated mode for acquisition of data for coronary artery calcification analysis
25.24	TAVI	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for TAVI scan for evaluation of coronary arteries intended for heart rates 30-74 bpm and the entire aorta.
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40	46.93	1255.42	25.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm and helical scan for the entire aorta.
			Helical	LargeBody	19.37 0.969	0.35	0.625	20	120	SmartmA 50-400mA NI=22 Avg mA=150	40	STND PLUS	SS40	5.05	261.70	49.94	Body 32	
25.26	SnapShot Segment		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 30-74 bpm
			Cardiac SEG	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSSEG	SS40:Slice	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 30-74 bpm
25.27	SnapShot Burst		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 75-113 bpm
			Cardiac SSB	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSB	SS40:Slice	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 75-113 bpm

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
25.28	SnapShot Burst Plus		Helical	LargeBody	55.00 1.375	0.6	5	40	120	40	0	STND PLUS	NONE	1.52	14.85	5.00	Body 32	Localizer scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	59.41	59.41	0	Body 32	Timing bolus scan for retrospectively gated Cardiac helical scan for evaluation of coronary arteries intended for heart rates 114+ bpm
			Cardiac SSBP	CardiacLarge	2.80 0.14	0.35	0.625	20	120	ECG 150-270mA Phase 70-80 Avg mA=270	0	STND SSB	SS40:Silence	47.63	559.64	10.00	Body 32	Retrospectively gated Cardiac helical scan with ECG modulated mA for evaluation of coronary arteries intended for heart rates 114+ bpm

Abdomen

Table 16-6

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.1	Abdomen Pelvis 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with Smart mA, Evaluation for abnormalities of abdominal and pelvic structures
26.2	Abdomen Pelvis 5mm SmartmA/ MAR		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Basic Abdomen Pelvis (0.8sec) with SmartmA and SmartMAR, Evaluation for abnormalities of abdominal and pelvic structures
26.3	Abdomen Pelvis + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	120	SmartmA 50-400mA NI=18 Avg mA=230	40	STND PLUS	SS40:Sli ce	17.83	746.78	40.00	Body 32	Basic Abdomen/Pelvis (0.8sec) with Smart mA, SmartPrep and DMPR, Evaluation for abnormalities for abdominal and pelvic structures
26.5	Renal Stone 5mm		Helical	LargeBody	39.37 0.984	0.8	5	40	120	130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan, Evaluation of kidney, ureters and bladder for renal stones
26.6	Renal Stone 5mm SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=130	40	STND PLUS	SS40:Sli ce	9.20	400.68	40.00	Body 32	Helical scan with SmartmA, Evaluation of kidney, ureters and bladder for renal stones
26.8	AAA + DMPR 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for helical scan with DMPR, Evaluation for abdominal aortic aneurysm
		DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	125	40	STND PLUS	SS40:Sli ce	13.99	585.51	40.00	Body 32	Helical scan with DMPR, Evaluation for abdominal aortic aneurysm
26.9	AAA + DMPR 0.625mm SmartmA/ SmartPrep	DMPR	Helical	LargeBody	19.37 0.969	0.8	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=125	40	STND PLUS	SS40:Sli ce	13.99	585.51	40.00	Body 32	Helical scan with DMPR, Smart mA and SmartPrep, Evaluation for abdominal aortic aneurysm
26.11	Dual Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Sli ce	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.12	Tri Phase Liver		Helical	LargeBody	39.37 0.984	0.8	5	40	120	95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
		MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing Bolus for Multi-group acquisition for early arterial, late arterial and portal venous phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition for portal venous phase of the liver.
26.13	Dual Phase Pancreas SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	225.56	30.00	Body 32	Non contrast evaluation of upper abdominal structures including pancreas
			Helical	LargeBody	39.37 0.984	0.6	2.5	40	120	SmartmA 50-400mA NI=15 Avg mA=180	40	STND PLUS	SS40:Slice	9.56	129.47	10.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial phase of the pancreas.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	267.16	30.00	Body 32	Acquisition with SmartmA and SmartPrep for portal venous phase of the pancreas
26.14	Dual Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 80-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.15	Tri Phase Liver SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=95	40	STND PLUS	SS40:Slice	6.72	292.81	40.00	Body 32	Non contrast evaluation of upper abdominal structures including liver
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for early arterial, phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	187.53	20.00	Body 32	Acquisition with SmartmA and SmartPrep for late arterial phase of the liver.
			Helical	LargeBody	39.37 0.984	0.6	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	7.96	346.80	40.00	Body 32	Acquisition with SmartmA and SmartPrep portal venous phase of the liver
26.19	Runoff 0.625mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	200	40	STND PLUS	SS40:Slice	16.91	1722.75	100.00	Body 32	Helical scan mode 0.625mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.20	Runoff 1.25mm	MIROI	Axial	LargeBody	1i	1	10 [0]	10	120	40	0	STND	NONE	50.92	50.92	0	Body 32	Timing bolus scan for evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
			Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)
26.21	Runoff 0.625mm SmartmA/ SmartPrep		Helical	LargeBody	19.37 0.969	0.6	0.625	20	140	SmartmA 50-340mA NI=22.1 Avg mA=200	40	STND PLUS	SS40:Slice	16.91	1722.75	100.00	Body 32	Helical scan mode 0.625mm with SmartmA and SmartPrep, Evaluation of vascular structures of abdomen, femurs and lower extremities (Runoff)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
26.22	Runoff 1.25mm SmartmA/ SmartPrep		Helical	LargeBody	39.37 0.984	0.6	1.25	40	140	SmartmA 50-340mA NI=21.45 Avg mA=150	40	STND PLUS	SS40:Slice	11.58	1198.98	100.00	Body 32	Helical scan mode 1.25mm with SmartmA and SmartPrep, Evaluation of vasacular structures of abdomen, femurs and lower extremities (Runoff)
26.24	CT Colonography 1.25mm	Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Supine acquisition for using Advantage CTC Pro for evaluation of the colon
		Advantage CTC Pro	Helical	LargeBody	39.37 0.984	0.6	1.25 [0.625]	40	120	70	40	STND PLUS	SS40:Slice	3.72	161.84	40.00	Body 32	Helical scan mode Prone acquisition for using Advantage CTC Pro for evaluation of the colon

L-Spine

Table 16-7

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
27.1	L-Spine 3 Level Axial		Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	120	95	40	STND	SS40:Slice	17.71	35.41	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
			Axial	LargeBody	8i	2	2.5	20	140	110	40	STND	SS40:Slice	29.81	59.62	1.75	Body 32	Axial scan mode, Evaluation of soft tissue and bony structures lumbar spine
27.3	L-Spine Survey Helical 2.5mm SmartmA		Helical	LargeBody	20.62 0.516	0.8	2.5	40	140	SmartmA 50-340mA NI=14 Avg mA=155	40	BONE PLUS	SS40:Slice	30.46	637.58	17.50	Body 32	Helical scan mode 2.5mm (0.8sec), Limited evaluation of lumbar spine
27.5	L-Spine Helical 0.625mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-340mA NI=20 Avg mA=205	40	BONE PLUS IQE+	SS40:Slice	52.68	1013.18	17.50	Body 32	Helical scan mode 0.625mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.6	L-Spine Helical 1.25mm + DMPR/ SmartmA	DMPR	Helical	LargeBody	20.62 0.516	1	1.25	40	140	SmartmA 50-340mA NI=18 Avg mA=110	40	BONE PLUS IQE	SS40:Slice	27.02	565.50	17.50	Body 32	Helical scan mode 1.25mm with DMPR and SmartmA, Evaluation of soft tissue and bony structures of lumbar spine
27.7	L-Spine Helical 0.625mm + DMPR/ SmartmA/ MAR	DMPR	Helical	LargeBody	10.62 0.531	1	0.625	20	140	SmartmA 50-340mA NI=20 Avg mA=205	40	BONE PLUS IQE+	SS40:Slice	52.68	1013.18	17.50	Body 32	Helical scan mode 0.625mm with DMPR, SmartmA and SmartMAR, Evaluation of soft tissue and bony structures of lumbar spine

Pelvis

Table 16-8

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
28.1	Pelvis for Fracture + DMPR/ SmartmA	DMPR	Helical	LargeBody	39.37 0.984	1	1.25	40	120	SmartmA 50-400mA NI=20 Avg mA=200	40	BONE PLUS IQE	SS40:Slice	17.70	241.74	10.13	Body 32	Helical scan with DMPR and SmartmA, Evaluation of pelvis bony structures for fracture
28.3	Pelvis Soft Tissue SmartmA		Helical	LargeBody	39.37 0.984	0.8	5	40	120	SmartmA 50-400mA NI=12 Avg mA=150	40	STND PLUS	SS40:Slice	10.62	303.06	25.00	Body 32	Helical scan with SmartmA, Evaluation of soft tissue structures of pelvic area for abnormality

Lower Extremity

Table 16-9

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
29.1	Lower Extremity Survey 2.5mm		Helical	LargeBody	20.62 0.516	1	2.5	40	120	75	40	STND PLUS	SS40:Slice	12.67	119.48	6.00	Body 32	Helical scan mode 2.5mm, Limited evaluation of lower extremity for abnormalities
29.3	Knee 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	95	40	STND PLUS IQE+	SS40:Slice	16.67	86.95	3.50	Body 32	Helical scan mode 0.625, Evaluation of soft tissue and bony anatomy of both knees
29.5	Ankle 0.625mm Single Joint		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.6	Ankle 0.625mm Both Joints		Helical	LargeBody	10.62 0.531	1	0.625	20	120	75	40	STND PLUS IQE+	SS40:Slice	13.16	68.65	3.50	Body 32	Helical scan mode 0.625mm, Evaluation of soft tissue and bony anatomy of the ankle area
29.7	Ankle 0.625mm Single Joint MAR		Helical	LargeBody	10.62 0.531	1	0.625	20	120	50	30	STND PLUS IQE+	SS30:Slice	8.77	45.76	3.50	Body 32	Helical scan mode 0.625mm with SmartMAR, Evaluation of soft tissue and bony anatomy of the ankle area

Reference Pediatric Protocols 32

Slice System, 55kW with 0.35sec/ ASiR

Head

Table 17-1

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.
For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
31.1	PED HEAD TO 18 Months		Axial	Ped Head	4i	0.5	5	20	120	170	0	STND	SS30:Slice	15.22	182.63	11.50	Head 16	Routine head (0.5sec) for infant (up to 18 months) 5mm
31.2	PED HEAD 18 MOS TO 5YRS		Axial	Head	4i	0.5	5	20	120	235	0	STND	SS30:Slice	21.17	254.02	11.50	Head 16	Routine head (0.5sec) for children (18 months to 5 years) 5mm
31.3	PED HEAD TO 5YRS TO 18YRS		Axial	Head	4i	1	5	20	120	125	0	STND	SS30:Slice	22.38	268.57	11.50	Head 16	Routine head (1.0sec) for children (5 years to 18 years) 5mm

Orbit

Table 17-2

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
32.1	Sinus Supine Helical + DMPR	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	50	0	BONE PLUS IQE+	SS20:Slice	8.43	77.18	7.44	Head 16	Routine Helical scan mode with DMPR, Evaluation of sinus area for abnormality in pediatric patient
32.3	Orbits Helical	DMPR	Helical	Head	10.62 0.531	0.5	0.625	20	120	105	0	STND PLUS IQE+	SS30:Slice	17.69	139.95	6.19	Head 16	Helical scan mode, Evaluation of orbital and face structures for abnormality in pediatric patient
32.5	Helical IAC 0.625mm Axial mode - Use Reformat for Coronal	Reformat	Helical	Head	10.62 0.531	0.5	0.625	20	120	100	0	STND PLUS IQE+	SS30:Slice	16.85	94.32	3.88	Head 16	Helical scan mode, Evaluation of inner ear structures for abnormality in pediatric patient

Chest

Table 17-3

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.1.1	RC 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	55	0	STND PLUS	SS30:Slice	1.30	13.07	7.50	Body 32	Weight and height based routine chest protocol
35.1.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSSEG	SS30:Slice	10.98	101.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.1.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 40-80 Avg mA=195	0	STND SSB	SS30:Slice	10.98	101.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.1.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	55	0	STND PLUS	SS30:Slice	0.61	5.98	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-195mA Phase 35-85 Avg mA=195	0	STND SSB	SS30:Slice	11.87	109.77	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.1.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-200mA NI=18 Avg mA=135	30	STND PLUS	SS30:Slice	1.52	14.25	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.2.1	RC 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	85	0	STND PLUS	SS30:Slice	2.02	20.20	7.50	Body 32	Weight and height based routine chest protocol
35.2.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSSEG	SS30:Slice	11.78	108.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.2.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 40-80 Avg mA=210	0	STND SSB	SS30:Slice	11.78	108.95	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.2.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 55-210mA Phase 35-85 Avg mA=210	0	STND SSB	SS30:Slice	12.76	118.06	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.2.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-220mA NI=18 Avg mA=150	30	STND PLUS	SS30:Slice	1.69	15.83	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.3.1	RC 9.5-11.5kg (20.9-25.4lbs)		Helical	SmallBody	27.50 1.375	0.6	3.75	20	120	70	0	STND PLUS	SS30:Slice	2.49	25.86	7.88	Body 32	Weight and height based routine chest protocol
35.3.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSSEG	SS30:Slice	12.75	117.92	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.3.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 40-80 Avg mA=225	0	STND SSB	SS30:Slice	12.75	117.92	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.3.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-225mA Phase 35-85 Avg mA=225	0	STND SSB	SS30:Slice	13.73	127.02	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.3.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-240mA NI=18 Avg mA=165	30	STND PLUS	SS30:Slice	1.86	17.41	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.4.1	RC 11.5-14.5kg (25.4-32.0lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	65	0	STND PLUS	SS30:Slice	2.31	26.61	9.00	Body 32	Weight and height based routine chest protocol
35.4.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSSEG	SS30:Slice	13.55	125.37	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.4.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 40-80 Avg mA=240	0	STND SSB	SS30:Slice	13.55	125.37	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.4.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-240mA Phase 35-85 Avg mA=240	0	STND SSB	SS30:Slice	14.63	135.36	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.4.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-280mA NI=18 Avg mA=170	30	STND PLUS	SS30:Slice	1.91	17.94	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.5.1	RC 14.5-18.5kg (32.0-40.8lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	70	0	STND PLUS	SS30:Slice	2.49	28.66	9.00	Body 32	Weight and height based routine chest protocol
35.5.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	14.89	137.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.5.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	14.89	137.70	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.5.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	16.13	149.16	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.5.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 50-300mA NI=18 Avg mA=205	30	STND PLUS	SS30:Slice	2.31	21.63	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.6.1	RC 18.5-22.5kg (40.8-49.6lbs)		Helical	SmallBody	27.50 1.375	0.6	5	20	120	75	0	STND PLUS	SS30:Slice	2.67	30.70	9.00	Body 32	Weight and height based routine chest protocol
35.6.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSSEG	SS30:Slice	15.70	145.20	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.6.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy-cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 40-80 Avg mA=280	0	STND SSB	SS30:Slice	15.70	145.20	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.6.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	80	85	0	STND PLUS	SS30:Slice	0.95	9.24	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	80	50	0	STND	NONE	21.66	10.83	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	80	ECG 70-280mA Phase 35-85 Avg mA=280	0	STND SSB	SS30:Slice	17.03	157.54	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.6.7	Non ECG Gated		Helical	PedBody	19.37 0.969	0.4	0.625	20	80	SmartmA 100-340mA NI=18 Avg mA=235	30	STND PLUS	SS30:Slice	2.65	24.80	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.7.1	RC 22.5-31.5kg(49.6-69.5lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	75	0	STND PLUS	SS30:Slice	2.85	42.10	10.00	Body 32	Weight and height based routine chest protocol
35.7.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSSEG	SS30:Slice	26.13	241.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.7.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 40-80 Avg mA=245	0	STND SSB	SS30:Slice	26.13	241.69	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.7.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 55-245mA Phase 35-85 Avg mA=245	0	STND SSB	SS30:Slice	28.44	263.09	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy·cm) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
35.7.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-360mA NI=18 Avg mA=185	30	STND PLUS	SS30:Slice	4.96	46.41	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.8.1	RC 31.5-40.5kg(69.5-89.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	85	0	STND PLUS	SS30:Slice	3.23	54.18	12.00	Body 32	Weight and height based routine chest protocol
35.8.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
			Cardiac SEG	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSSEG	SS30:Slice	28.48	263.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.8.4	SnapShot Burst		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 40-80 Avg mA=265	0	STND SSB	SS30:Slice	28.48	263.44	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.8.5	SnapShot Burst+		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacSmall	2.80 0.14	0.35	0.625	20	100	ECG 70-265mA Phase 35-85 Avg mA=265	0	STND SSB	SS30:Slice	30.86	285.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.8.7	Non ECG Gated		Helical	SmallBody	19.37 0.969	0.5	0.625	20	100	SmartmA 100-400mA NI=16 Avg mA=275	30	STND PLUS	SS30:Slice	7.42	69.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA
35.9.1	RC 40.5-55.0kg(89.3-121.3lbs)		Helical	LargeBody	55.00 1.375	0.6	5	40	120	90	0	STND PLUS	SS30:Slice	3.42	67.63	15.00	Body 32	Weight and height based routine chest protocol
35.9.3	SnapShot Segment		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Slice	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBody	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
			Cardiac SEG	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSSEG	SS30:Si ce	32.69	302.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (30-74 BPM)
35.9.4	SnapShot Burst		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Si ce	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
			Cardiac SSB	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 120-295mA Phase 40-80 Avg mA=295	0	STND SSB	SS30:Si ce	32.69	302.41	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (75-104 BPM)
35.9.5	SnapShot Burst+		Helical	SmallBo dy	55.00 1.375	0.6	2.5	40	100	85	0	STND PLUS	SS30:Si ce	1.80	17.58	5.00	Body 32	Weight and height based protocol for localization
		MIROI	Axial	SmallBo dy	1i	1	5 [0]	5	100	50	0	STND	NONE	41.21	20.60	0	Body 32	Timing bolus for Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
			Cardiac SSBP	CardiacS mall	2.80 0.14	0.35	0.625	20	100	ECG 120-295mA Phase 35-85 Avg mA=295	0	STND SSB	SS30:Si ce	34.82	322.13	7.50	Body 32	Weight and height based cardiac protocol with ECG modulated mA (> 105 BPM)
35.9.7	Non ECG Gated		Helical	SmallBo dy	19.37 0.969	0.5	0.625	20	100	SmartmA 100-440mA NI=16 Avg mA=275	30	STND PLUS	SS30:Si ce	7.42	69.62	7.50	Body 32	Weight and height based non ECG gated protocol for cardiac with SmartmA

Abdomen

Table 17-4

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (second s)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·c m) ASiR	Scan Length (cm)	Phantom (cm)	Description
36.1.1	Abd 6.0-7.5kg (13.2-16.5lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	70	0	STND PLUS	SS30:Sli ce	1.66	17.26	7.88	Body 32	Weight and height based routine abdomen protocol
36.2.1	Abd 7.5-9.5kg (16.5-20.9lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	110	0	STND PLUS	SS30:Sli ce	2.61	30.05	9.00	Body 32	Weight and height based routine abdomen protocol
36.3.1	Abd 9.5- 11.5kg (20.9- 25.4lbs)		Helical	PedBody	27.50 1.375	0.4	3.75	20	120	125	0	STND PLUS	SS30:Sli ce	2.96	37.48	10.13	Body 32	Weight and height based routine abdomen protocol
36.4.1	Abd 11.5- 14.5kg (25.4- 32.0lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.52	34.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.5.1	Abd 14.5- 18.5kg (32.0- 40.8lbs)		Helical	SmallBo dy	27.50 1.375	0.5	5	20	120	95	0	STND PLUS	SS30:Sli ce	2.82	38.05	11.00	Body 32	Weight and height based routine abdomen protocol
36.6.1	Abd 18.5- 22.5kg (40.8- 49.6lbs)		Helical	LargeBo dy	27.50 1.375	0.5	5	20	120	85	0	STND PLUS	SS30:Sli ce	2.88	44.70	13.00	Body 32	Weight and height based routine abdomen protocol
36.7.1	Abd 22.5- 31.5kg(49.6- 69.5lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	105	0	STND PLUS	SS30:Sli ce	3.33	65.78	15.00	Body 32	Weight and height based routine abdomen protocol
36.8.1	Abd 31.5- 40.5kg(69.5- 89.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	120	0	STND PLUS	SS30:Sli ce	3.80	75.17	15.00	Body 32	Weight and height based routine abdomen protocol
36.9.1	Abd 40.5- 55.0kg(89.3- 121.3lbs)		Helical	LargeBo dy	55.00 1.375	0.5	5	40	120	140	0	STND PLUS	SS30:Sli ce	4.43	109.87	20.00	Body 32	Weight and height based routine abdomen protocol

Miscellaneous

Table 17-5

[Interval]*- For Helical acquisitions, Interval unless otherwise indicated is equal to the slice thickness.

For Axial and Cine acquisitions, Interval unless otherwise indicated is equal to the Beam Collimation/Detector Coverage.

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimation (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.1	Pink Zone Routine 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 65-130mA NI=5 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.2	Pink Zone Routine 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 80-160mA NI=7 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.3	Pink Zone Routine 61-100 lbs/27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 95-190mA NI=10 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.4	Pink Zone Routine 101-200 lbs/45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 110-220mA NI=12 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.5	Pink Zone Routine >200 lbs/>90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 125-300mA NI=15 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Routine procedure based protocols
40.7	Green Zone Lower Dose 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	5	40	80	SmartmA 50-100mA NI=7 Avg mA=60	0	STND FULL	NONE	0.67	9.52	9.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.8	Green Zone Lower Dose 21-60 lbs/9.1-27.2 kg		Helical	PedBody	55.00 1.375	0.6	5	40	100	SmartmA 60-120mA NI=9 Avg mA=60	0	STND FULL	NONE	1.27	30.82	19.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols

Protocol Number	Protocol Name	Post Process	Scan Type	SFOV	Pitch Table Speed Row	Rotation Time (seconds)	Slice Thickness (mm) [Interval]* (mm)	Beam Collimat ion (mm)	kV	mA Min-Max NI Avg mA	Dose Reduction (%)	Recon Type	ASiR (%)	CTDI vol (mGy) ASiR	DLP (mGy·cm) ASiR	Scan Length (cm)	Phantom (cm)	Description
40.9	Green Zone Lower Dose 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 70-140mA NI=11 Avg mA=60	0	STND FULL	NONE	2.28	78.15	29.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.10	Green Zone Lower Dose 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 80-160mA NI=13 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.11	Green Zone Lower Dose >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	5	40	120	SmartmA 90-240mA NI=16 Avg mA=60	0	STND FULL	NONE	2.28	100.95	39.50	Body 32	Pediatric and Small patient Lower Dose procedure based protocols
40.13	Grey Zone CT Angio 0-20 lbs/0-9 kg		Helical	PedBody	55.00 1.375	0.6	2.5	40	80	SmartmA 100-200mA NI=5 Avg mA=120	0	STND FULL	NONE	1.33	19.38	9.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.14	Grey Zone CT Angio 21-60 lbs/9.1-27.2 kg		Helical	SmallBody	55.00 1.375	0.6	2.5	40	100	SmartmA 120-240mA NI=8 Avg mA=120	0	STND FULL	NONE	2.54	62.28	19.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.15	Grey Zone CT Angio 61-100 lbs/ 27.3-45.4 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=10 Avg mA=120	0	STND FULL	NONE	4.56	157.44	29.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.16	Grey Zone CT Angio 101-200 lbs/ 45.4-90.7 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-240mA NI=12 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols
40.17	Grey Zone CT Angio >200 lbs/ >90.8 kg		Helical	LargeBody	55.00 1.375	0.6	2.5	40	120	SmartmA 120-300mA NI=15 Avg mA=120	0	STND FULL	NONE	4.56	203.05	39.75	Body 32	Pediatric and Small patient Angio procedure based protocols

Lung Cancer Screening

Table 18-1 Lung Cancer Screening protocol Manual mA

Parameters	Series 1	Series 2 PMR 1	Series 3 PMR 2	Series 4 PMR3 Optional	Series 5 PMR 4 Optional
Scan Type	Scout	NA	NA	NA	NA
Scout Plane	90 and 0	NA	NA	NA	NA
Start/End Location	S60-I300	NA	NA	NA	NA
kV	120	NA	NA	NA	NA
mA	10	NA	NA	NA	NA
AutoVoice	2	NA	NA	NA	NA
WW/WL	500/100	NA	NA	NA	NA
Scan Type	NA	Helical	NA	NA	NA
Rotation Time (s)	NA	0.5	NA	NA	NA
Start/End Location	NA	S0-I250	D	D	D
Thickness Speed					
Detector Coverage	NA	40 64 x0.625 * 32 x1.25	NA	NA	NA
Slice Thickness	NA	2.5	1.25	2.5	1.25
Pitch mm/rot	NA	0.984:1 39.375 mm/rot	NA	NA	NA
Rotation Time (s)	NA	0.5	NA	NA	NA
Interval (mm)	NA	2	0.625	2	0.625
Gantry Tilt	NA	0	NA	NA	NA
SFOV	NA	Large Body	NA	NA	NA
kV	NA	120	NA	NA	NA
mA	NA	Small - 30 Medium - 50 Large - 80	NA	NA	NA
Timing					
Prep Group	NA	AutoVoice 2	NA	NA	NA
ISD	NA	NA	NA	NA	NA
Breath Hold	NA	0	NA	NA	NA
Breathe Time	NA	0	NA	NA	NA
Voice Lights Timer	NA	2 On On	NA	NA	NA
Recon					
DFOV (cm)	NA	36	D	D	D
R/L Center (mm)	NA	S0	D	D	D
A/P Center (mm)	NA	A0	D	D	D
Recon Type	Standard	Standard	Standard	Bone	Bone
Matrix	512	512	512	512	512
Recon Options					
Recon Mode	NA	Full	Full	Full	Full
IQ Enhance	NA	Off	Off	Off	Off
WW/WL	NA	350/35	350/35	1500/-700	1500/-700
ASiR Setup	NA	Optional	Optional	Optional	Optional
Cardiac Filter	NA	NA	NA	NA	NA
Temporal Enhance	NA	NA	NA	NA	NA

Parameters	Series 1	Series 2 PMR 1	Series 3 PMR 2	Series 4 PMR3 Optional	Series 5 PMR 4 Optional
Auto Apps	NA	Off	Optional DMPR Coronal, Sagittal MIP	Off	Optional DMPR Coronal, Sagittal MIP
CDTI _{vol} (mGy)	NA	Small – 1.15 Medium – 1.92 Large – 3.07	NA	NA	NA
DLP (mGy-cm)	NA	Small – 34.11 Medium – 56.85 Large – 90.96	NA	NA	NA
Phantom (cm)	NA	Body 32	NA	NA	NA
Notification Value	NA	Factor: 2	NA	NA	NA
Filming	NA	Available	Available	Available	Available

Lexicon

Lexicon

For the CT technologist who operates multiple scanner models, perhaps from multiple manufacturers, the variability in names for important scan acquisition and reconstruction parameters can lead to confusion, reduced comfort and an increased potential for error. The intent of this CT terminology lexicon is to allow users to translate important CT acquisition and reconstruction terms between different manufacturers' systems.

The following tables provide the CT terminology for GE and is a result of the AAPM CT Terminology working group. It identifies relevant terms from established lexicon (e.g., Radlex and DICOM) and other relevant literature published.

The AAPM website will provide updates when changes in standardization of terminology occur. Visit www.aapm.org, search CT Protocols, and select Lexicon for full listing of terminology across various manufactures of CT equipment.

The generic descriptions or terms in the first column are intended to orient the user to the relevant concepts; they are not consensus "preferred terms." The generic descriptions are not based on any single existing or pending terminology standard; however the references cited below were consulted in developing the generic descriptions.

A number of individuals and groups have advocated for terminology standardization in CT, including at a March 30-31, 2010 FDA public meeting entitled "Device Improvements to Reduce Unnecessary Radiation Exposure from Medical Imaging" (transcripts available at the FDA website: www.fda.gov in Medical Devices). Participants proposed a cooperative effort among professional organizations (AAPM, ASRT, ACR, etc.), industry, and the FDA.

Scan Acquisition and User Interface Basics

Table 19-1 Scan acquisition and user interface terms

Generic Description	GE Name
The portion of the user interface where scans are prescribed	Exam Rx
Other portions of the user interface , such as where reconstructed images are viewed	Desktop
CT localizer radiograph (i.e. the scanned projection radiograph, often acquired by the CT system to allow the user to prescribe the start and end locations of the scan range)	Scout
Axial scan mode: Data acquisition while the patient table remains stationary; the table position may be incremented between X-ray exposures to collect data over a longer z axis range.	Axial
Helical or Spiral scan mode: Data acquisition while the patient table is continuously moving along the z axis.	Helical
Dynamic scan mode - single detector width: Data acquisition at multiple time points over the same anatomic location(s) while the patient table remains stationary; X-ray exposure can be continuous or intermittent	Cine or zero interval Axial
Dynamic scan mode - multiple detector widths: Data acquisition at multiple time points over the same anatomic location(s) while the patient table cycles back and forth between designated start and end locations in order image a region wider than the detector	Shuttle
Interventional CT - Intermittent X-ray exposures	SmartStep
Interventional CT - Continuous X-ray exposures	SmartView

Generic Description	GE Name
Table increment (mm) per 360 degree rotation of the X-ray tube (axial scan mode)	Interval
Table feed per 360 degree rotation of the X-ray tube (helical scan mode)	Speed (mm/rot)
Field of measurement: Diameter of the circular region within the scan plane over which projection data are collected. Nominally equal to the diameter of the primary beam at isocenter in the axial plane.	Scan Field of View (SFOV, cm)
Tube current: Number of electrons accelerated across an X-ray tube per unit time, expressed in units of milliamperere (mA)	mA
Tube current-time product: The product of tube current and exposure time per rotation, expressed in units of milliamperere • seconds (mAs). In axial scan mode, this is equal to tube current × (scan angle ÷ 360) × rotation time. In helical scan mode, this is equal to tube current × rotation time.	Not used on this system
Effective tube current-time product: In helical scan mode, this is the product of tube current and rotation time (expressed in units of milliamperere • seconds (mAs) ÷ pitch)	Not used on this system
Tube potential: The electric potential applied across an X-ray tube to accelerate electrons towards a target material, expressed in units of kilovolts (kV)	kV
Pitch: Unitless parameter used to describe the table travel during helical CT; equal to table travel (mm) per gantry rotation ÷ total nominal beam width (mm)	Pitch
Automated patient instructions	AutoVoice

Dose Modulation and Reduction Tools

Table 19-2 Dose modulation and reduction tool terms

Generic Description	GE Name
Automatic exposure control (AEC): A scanner feature that automatically adapts the X-ray tube current to the overall patient size to achieve a specified level of image quality	Available in AutomA and SmartmA
Angular tube current modulation	SmartScan (CT/i only)
Longitudinal tube current modulation	AutomA
Angular and longitudinal tube current modulation	SmartmA (x, y, z)
ECG-based tube current modulation	ECG Modulated mA
Image quality reference parameter for AEC	Noise Index

Multi-slice Detector Geometry

Table 19-3 Multi-slice detector geometry terms

Generic Description	GE Name
Multi-slice detector array design	Fixed
Detector configuration	Detector Configuration

Figure 19-1 Detector Coverage and Slice Thickness - Detector rows (N) and slice thickness (T) selection console screen image

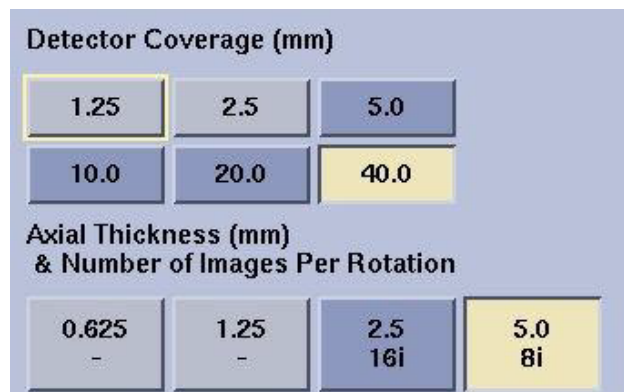


Image Reconstruction and Display

Table 19-4 Image reconstruction and display terms

Generic Description	GE Name
Window width: Range of CT numbers (maximum - minimum) that are distributed over the viewable grey scale of the display device or film	Window Width
Window center: The CT number in the center of the viewable grey scale	Window Level
Reconstruction field of view: Width of the square region mapped to the reconstructed image matrix	Display Field of View (DFOV) (cm)
Prescribing the reconstruction parameters prior to scan acquisition	Prospective recon
Prescribing the reconstruction parameters after scan acquisition	Retrospective recon
Reconstruction property that determines sharpness or smoothness of image in the axial plane	Algorithm
Helical interpolation options to achieve a wider or narrower section sensitivity profile	Full (narrower) or Plus (wider) mode
Nominal width of reconstructed image along the z axis	Thickness (mm)
Distance between two consecutive reconstructed images	Interval
Fast but lower-quality reconstructed images for rapid review of entire exam	Image Check
Off-center reconstruction coordinates are called	RL Center AP Center
Flip or rotate the image orientation is called	Flip/rotate
Image modifications to alter sharpness or smoothness (done in image space without reconstructing images)	Image Filters

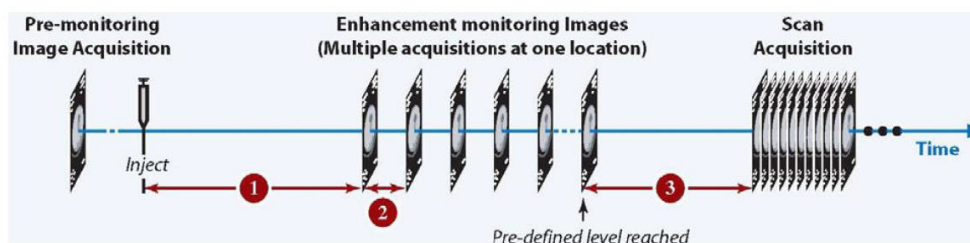
Contrast Media Tools

Table 19-5 Contrast media tool terms

Generic Description	GE Name
Bolus tracking: Scanner feature to automatically initiate a prescribed axial, helical or dynamic scan when a threshold level of contrast enhancement is reached at a specified region of interest	Smart Prep
Test Bolus: Scan mode used to measure the contrast transit time using a small injection of contrast media	Take axial scans at zero table feed and process with MIROI

Generic Description	GE Name
Time-attenuation curve (TAC): Graph of the contrast enhancement versus time	Smart Prep graph or MIROI graph
Threshold: CT number (HU) where bolus tracking tool will trigger the system to begin the scan	Transition ROI Threshold
Scanner feature used to quantitatively evaluate the TAC	MIROI (multiple image region of interest)
Monitoring delay: Time from injection to the start of monitoring scans (Time 1 in figure below)	Monitoring Delay
Monitoring interval: Time between consecutive monitoring scans to (Time 2 in figure below)	Monitor ISD (InterScan Delay)
Scan delay: Time from when threshold is reached and prescribed axial, helical or dynamic scan begins (Time 3 in figure below)	Diagnostic delay

Figure 19-2



Multi-planar Formats and 3D Processing

Table 19-6 Multi-planar formats and 3D processing terms

Generic Description	GE Name
Reformatted image at an oblique plane (not an axial, coronal, or sagittal)	Oblique reformat
Saving images at various viewing angles about a volume or surface rendered object	Batch Loop
Saving images at various planes through a volume	Batch Reformat
Surface-rendered object	3D
Volume-rendered object	Volume Rendered image (VR)

Service and Application Tools

Table 19-7 Service and application tool terms

Generic Description	GE Name
X-ray tube warm up	Tube Warm-up (tube warm up)
Daily calibrations	Fast Cals (done in daily prep)
Application information	Learning Solutions or User Manual
Application support assistance	Insite or Ilinq

Workflow

Table 19-8 Workflow terms

Generic Description	GE Name
Scheduled (but not yet scanned) patient list is called	Patient Schedule
Already scanned patient list is called	List/Select
User comments or text added to an image is called	User annotation
Filming tools are called	Auto/manual film composer
Data page summarizing scan parameters, CTDIvol and DLP	Exam Text Page or Series Text Page
Sorting patient list	Sort

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